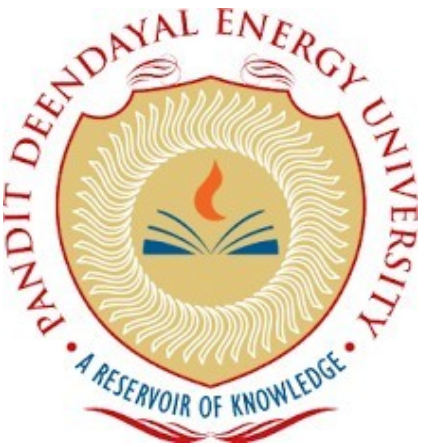
B.Tech Computer Engineering
Semester – VII

Agile Methodology & Devops Lab

20CP408P

ACY: 2026-27



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Lab 1

Aim:

Overview of Version Control Using Git

Theory

1. Version Control

Version Control is a system used to track and manage changes made to source code. It helps developers maintain different versions, recover previous changes, and collaborate on projects.

Advantages:

1. Tracks changes in source code.
2. Helps recover previous versions.
3. Supports teamwork and collaboration.
4. Maintains a history of modifications.

2. Introduction to Git

Git is a distributed version control system developed by Linus Torvalds in 2005. It is used to manage repositories, track changes, create branches, and collaborate on software projects.

3. Types of Version Control Systems

Centralized Version Control System (CVCS):

1. Uses a single central repository.
2. Requires network access for most operations.
3. Server failure may result in data loss.

Distributed Version Control System (DVCS):

1. Every developer has a complete repository copy.
2. Supports offline operations.

3. Provides faster operations and better data recovery.

4. Important Git Commands

- **git init:** Initializes a new Git repository in the current directory.
- **touch hello.txt:** Creates a new empty text file.
- **nano hello.txt:** Opens a file for editing in the terminal.
- **cat hello.txt:** Displays the contents of a file.
- **git status:** Shows the current status of files, including tracked, untracked, and staged changes.
- **git add .:** Adds all changes to the staging area before committing.
- **git commit -m "First commit":** Saves staged changes in the local repository with a message.
- **git remote add origin <url>:** Connects the local repository to a remote GitHub repository.
- **git push -u origin main:** Uploads local commits to the main branch of the remote repository.
- **git branch Naitik:** Creates a new branch named Naitik.
- **git checkout Naitik:** Switches from the current branch to the Naitik branch.
- **git push -u origin branch2:** Uploads the new branch and its commits to GitHub.
- **git clone <url>:** Creates a local copy of an existing remote repository.

5. Important Git Concepts

Repository: A storage location containing project files and their version history.

Staging Area: A temporary area where changes are prepared before committing.

Commit: A saved record of changes made to the repository.

Branch: An independent line of development used to work on different features.

Pull Request: A request to review and combine changes from one branch into another.

Merge: Combines changes from different branches into a single branch.

6. Git Workflow

1. Create a repository on GitHub.
2. Initialize the local repository using git init.
3. Create files and check their status using git status.
4. Stage changes using git add .
5. Save changes using git commit.
6. Connect the repository using git remote add origin.
7. Upload changes using git push.
8. Create and switch branches using git branch and git checkout.
9. Create a pull request and merge changes into main.
10. Clone the repository using git clone.

7. Result

Successfully created and managed a Git repository, performed basic Git commands, created branches, pushed changes to GitHub, and merged changes using a pull request.

Source Code:

file1.txt

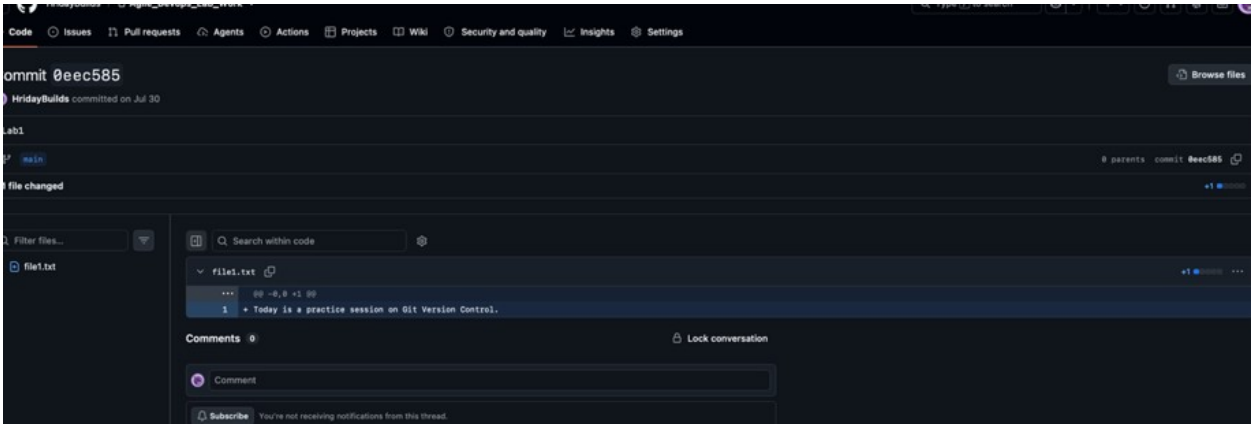
Today is a practice session on Git Version Control.

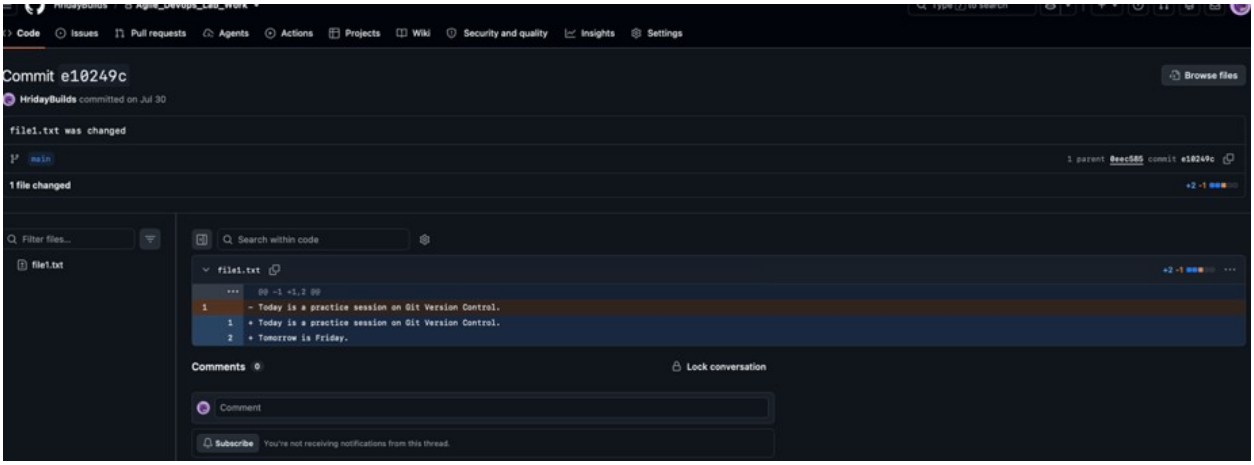
Tomorrow is Friday.

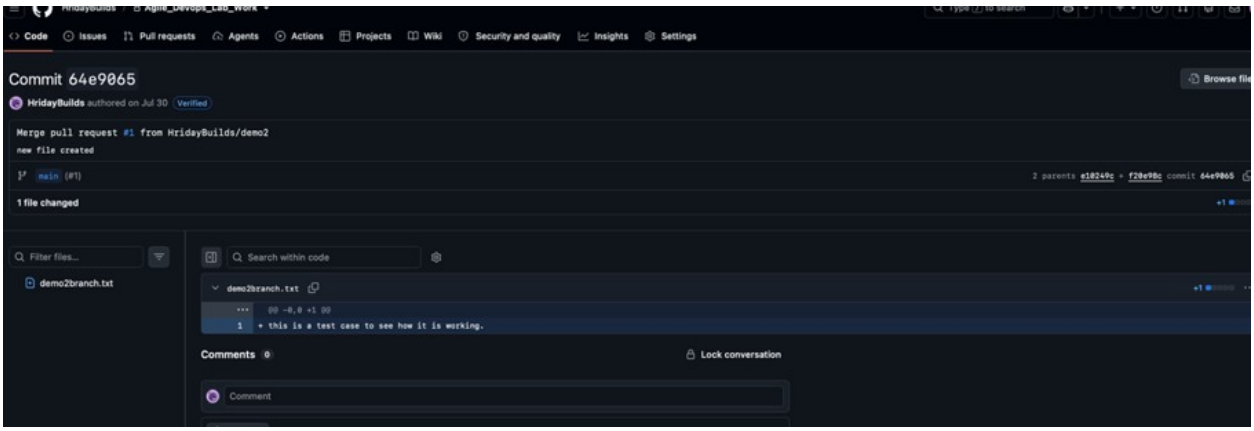
Demo2branch.txt

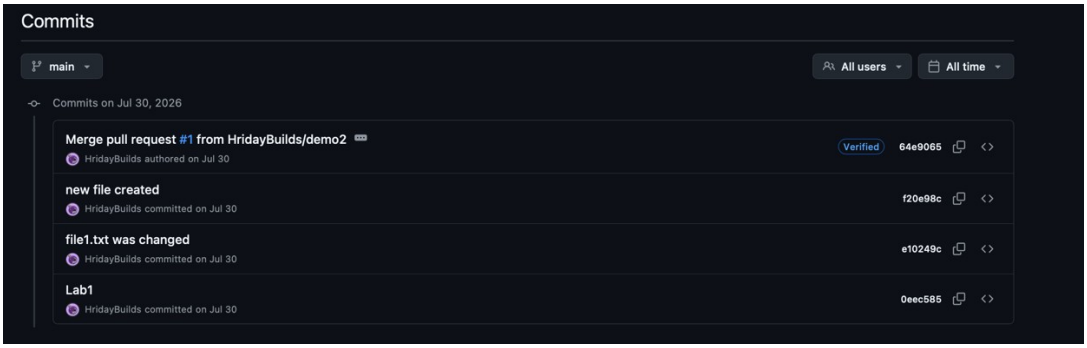
this is a test case to see how it is working.

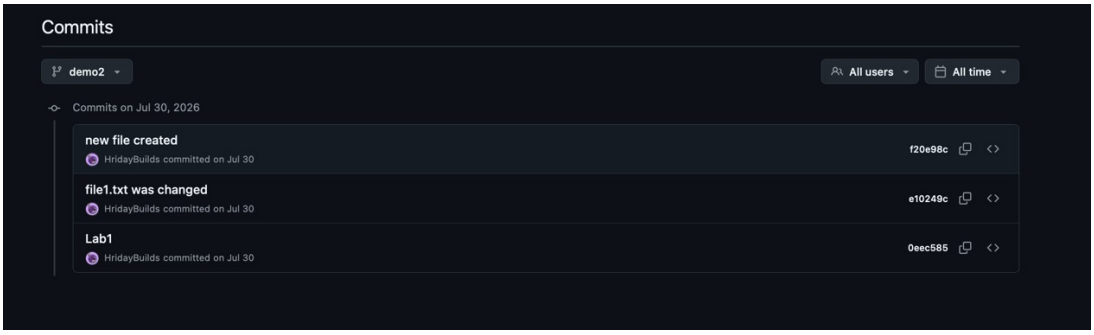
Output:

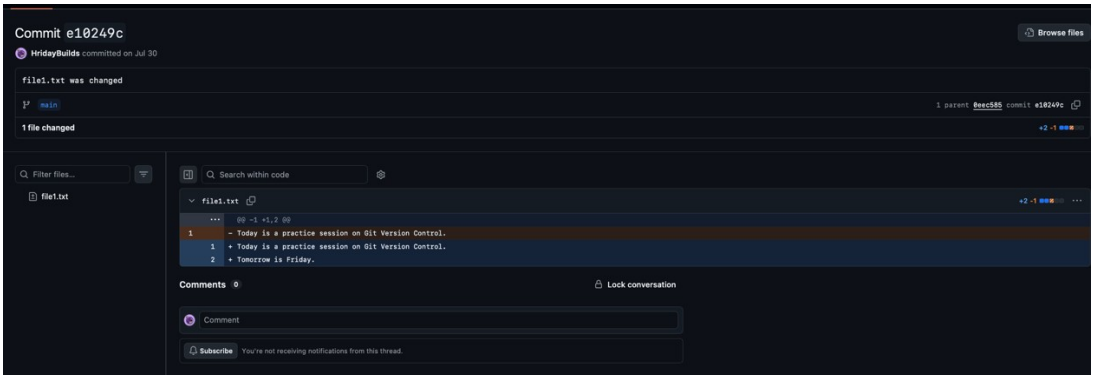

Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
ommit 0eec585
HridayBuilds committed on Jul 30
Browse files
lab1
main
0 parents commit 0eec585
file changed
Filter files...
file1.txt
Search within code
file1.txt
@@ -0,0 +1 @@
+ Today is a practice session on Git Version Control.
Comments 0
Lock conversation
Comment
Subscribe
You're not receiving notifications from this thread.


Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Commit e10249c
HridayBuilds committed on Jul 30
Browse files
file1.txt was changed
main
1 parent 0eec585 commit e10249c
1 file changed
Filter files...
file1.txt
Search within code
file1.txt
@@ -1 +1,2 @@
- Today is a practice session on Git Version Control.
+ Today is a practice session on Git Version Control.
+ Tomorrow is Friday.
Comments 0
Lock conversation
Comment
Subscribe
You're not receiving notifications from this thread.


Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Commit 64e9065
HridayBuilds authored on Jul 30
Verified
Merge pull request #1 from HridayBuilds/demo2
new file created
main (#1)
2 parents e10249c + f20e98c commit 64e9065
1 file changed
Filter files...
demo2branch.txt
Search within code
demo2branch.txt
@@ -0,0 +1 @@
+ this is a test case to see how it is working.
Comments 0
Lock conversation
Comment


Commits
main
All users
All time
Commits on Jul 30, 2026
Merge pull request #1 from HridayBuilds/demo2
HridayBuilds authored on Jul 30
Verified
64e9065
new file created
HridayBuilds committed on Jul 30
f20e98c
file1.txt was changed
HridayBuilds committed on Jul 30
e10249c
Lab1
HridayBuilds committed on Jul 30
0eec585


Commits
demo2
All users
All time
Commits on Jul 30, 2026
new file created
HridayBuilds committed on Jul 30
f20e98c
file1.txt was changed
HridayBuilds committed on Jul 30
e10249c
Lab1
HridayBuilds committed on Jul 30
0eec585


Commit e10249c
HridayBuilds committed on Jul 30
Browse files
file1.txt was changed
main
1 parent 0eec585 commit e10249c
1 file changed
+2 -1
Filter files...
file1.txt
Search within code
file1.txt
+2 -1
@@ -1 +1,2 @@
- Today is a practice session on Git Version Control.
+ Today is a practice session on Git Version Control.
+ Tomorrow is Friday.
Comments 0
Lock conversation
Comment
Subscribe
You're not receiving notifications from this thread.

Lab 2 and 3

Aim:

Creating infrastructure on AWS/Azure using Cloud Formation Templates and Using Intrinsic Functions in CloudFormation Templates

Theory:

1. Introduction to AWS

Amazon Web Services (AWS) is a cloud computing platform that provides services such as virtual servers, storage, networking, and databases. It allows users to run applications without maintaining physical hardware.

2. Amazon EC2

Amazon Elastic Compute Cloud (EC2) is an AWS service used to create and manage virtual servers called instances.

Features:

1. Provides virtual servers in the cloud.
2. Allows users to select CPU, memory, and storage.
3. Supports different operating systems.
4. Provides scalable computing resources.

3. AWS CloudFormation

AWS CloudFormation is a service used to create and manage AWS resources automatically using templates. It follows the concept of Infrastructure as Code (IaC), where cloud infrastructure is defined using configuration files instead of manually creating resources.

Advantages:

1. Automates resource creation.
2. Reduces manual configuration.
3. Provides consistent infrastructure.

4. Allows multiple resources to be managed together.

4. Important AWS Components

- **EC2 Instance:** A virtual server used to run applications in the AWS cloud.
- **Instance Type:** Defines the computing capacity of an instance. In this experiment, t3.micro is used.
- **Amazon Machine Image (AMI):** A predefined image containing the operating system required to launch an EC2 instance. Amazon Linux 2023 is used.
- **Virtual Private Cloud (VPC):** A logically isolated network within AWS where cloud resources are deployed.
- **Subnet:** A smaller network within a VPC where EC2 instances are launched.
- **Security Group:** A virtual firewall that controls incoming and outgoing network traffic.
- **Elastic IP:** A static public IPv4 address assigned to an EC2 instance for internet accessibility.
- **Private IP:** An internal IP address used for communication within the VPC.
- **EC2 Instance Connect:** A service that allows users to securely connect to an EC2 instance through the AWS console.

5. Security Group Configuration

Security groups control network access to EC2 instances using inbound and outbound rules.

In this experiment:

1. **Port 22 (SSH):** Allows remote access to the EC2 instance.
2. **Port 80 (HTTP):** Allows web traffic to reach the instance.
3. **0.0.0.0/0:** Allows access from any IPv4 address.

For better security, SSH access should be restricted to trusted IP addresses.

6. CloudFormation Template Components

- **Template Format Version:** Specifies the CloudFormation template format.
- **Description:** Explains the purpose of the template.
- **Parameters:** Accepts user inputs such as instance type, AMI, VPC, and subnet.
- **Resources:** Defines the AWS resources to be created, including the EC2 instance, security group, and Elastic IP.
- **Outputs:** Displays important information after deployment, such as instance ID, private IP, and public Elastic IP.

Source Code:

```
AWSTemplateFormatVersion: '2010-09-09'
Description: >
  Simple EC2 stack - Security Group + EC2 Instance (private IP automatic)
  + Elastic IP (public). You choose the VPC and Subnet from dropdowns
  (pick the ones labeled "default"). No Lambda, no SSM, no Key Pair needed
  - connect using EC2 Instance Connect.

Parameters:
  InstanceType:
    Description: EC2 instance type
    Type: String
    Default: t3.micro

  LatestAmiId:
    Description: AMI ID (defaults to latest Amazon Linux 2023 via SSM)
    Type: AWS::SSM::Parameter::Value<AWS::EC2::Image::Id>
    Default: /aws/service/ami-amazon-linux-latest/al2023-ami-kernel-default-x86_64

  VpcId:
    Description: Select your DEFAULT VPC from the dropdown
    Type: AWS::EC2::VPC::Id

  SubnetId:
    Description: Select a DEFAULT subnet (inside the VPC above) from the dropdown
    Type: AWS::EC2::Subnet::Id

Resources:

  # ---------- Security Group ----------
  InstanceSecurityGroup:
    Type: AWS::EC2::SecurityGroup
```

```
  Properties:
    GroupDescription: Allow SSH (for EC2 Instance Connect) and HTTP inbound
    VpcId: !Ref VpcId
    SecurityGroupIngress:
      - IpProtocol: tcp
        FromPort: 22
        ToPort: 22
        CidrIp: 0.0.0.0/0
      - IpProtocol: tcp
        FromPort: 80
        ToPort: 80
        CidrIp: 0.0.0.0/0
    Tags:
      - Key: Name
        Value: AgileLab-SG

 # ---------- EC2 Instance ----------
 MyEC2Instance:
   Type: AWS::EC2::Instance
   Properties:
     InstanceType: !Ref InstanceType
     ImageId: !Ref LatestAmiId
     SubnetId: !Ref SubnetId
     SecurityGroupIds:
       - !Ref InstanceSecurityGroup
     Tags:
       - Key: Name
         Value: AgileLab-Instance

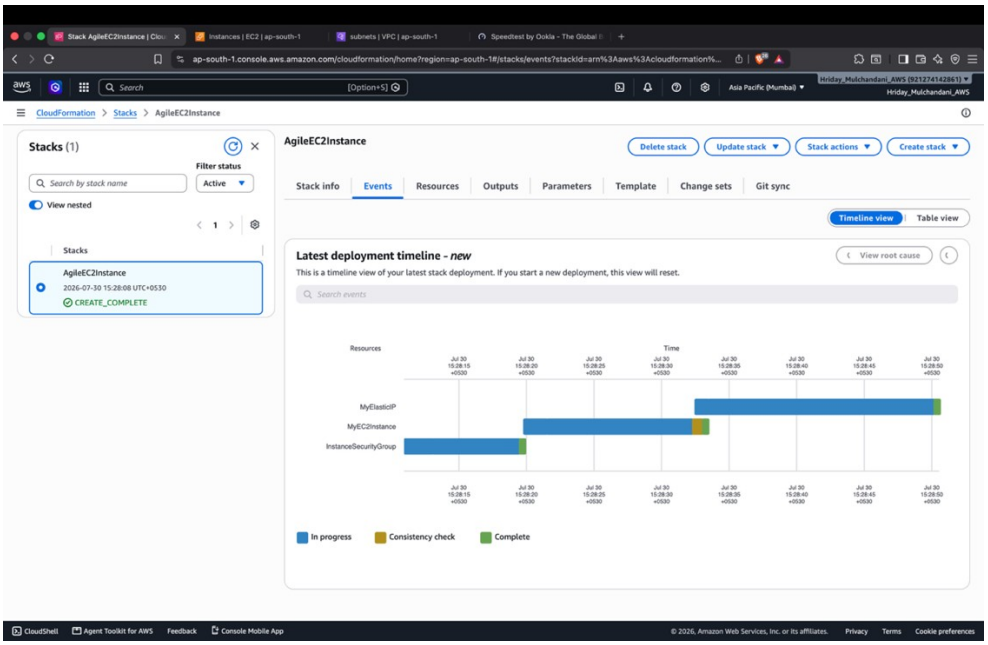
 # ---------- Elastic IP ----------
 MyElasticIP:
   Type: AWS::EC2::EIP
   Properties:
     Domain: vpc
     InstanceId: !Ref MyEC2Instance
     Tags:
       - Key: Name
         Value: AgileLab-EIP

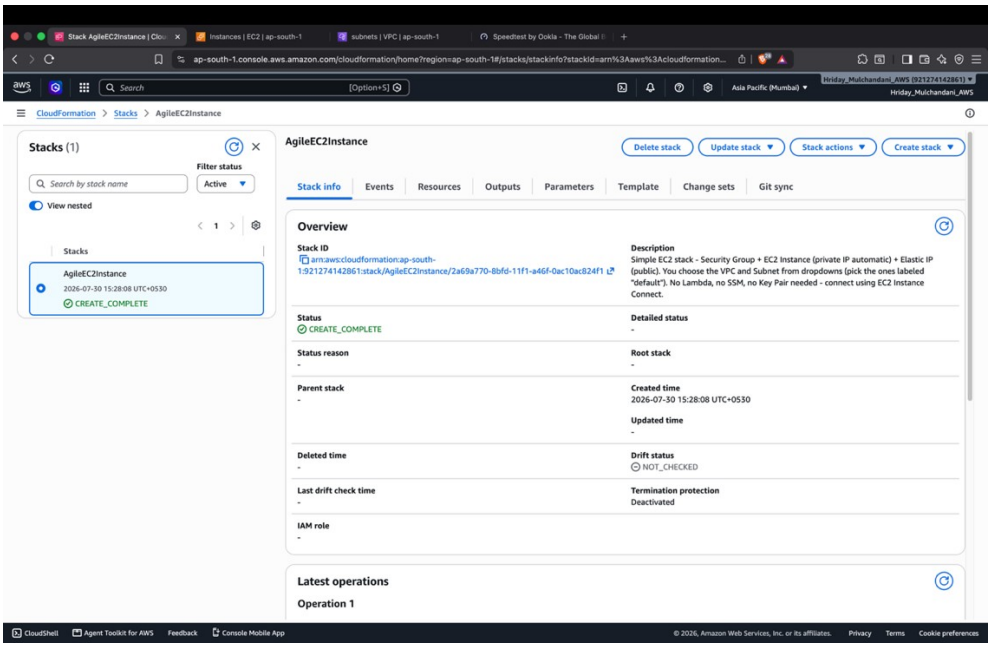
Outputs:
 InstanceId:
   Description: EC2 Instance ID
   Value: !Ref MyEC2Instance

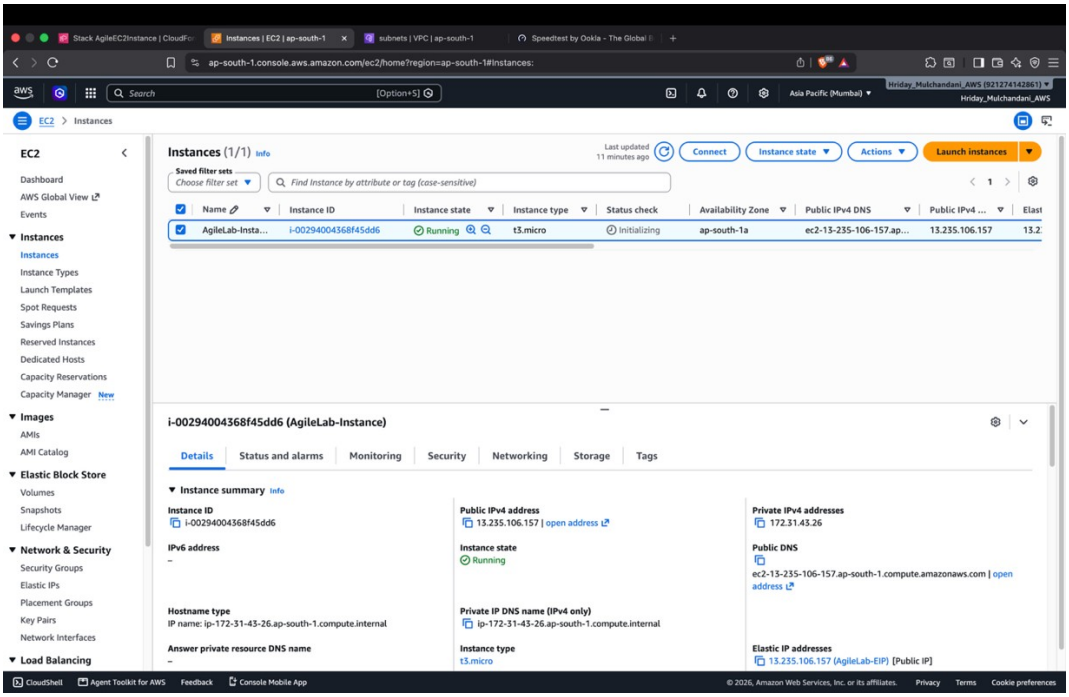
 PrivateIP:
   Description: Private IP address
   Value: !GetAtt MyEC2Instance.PrivateIp
```

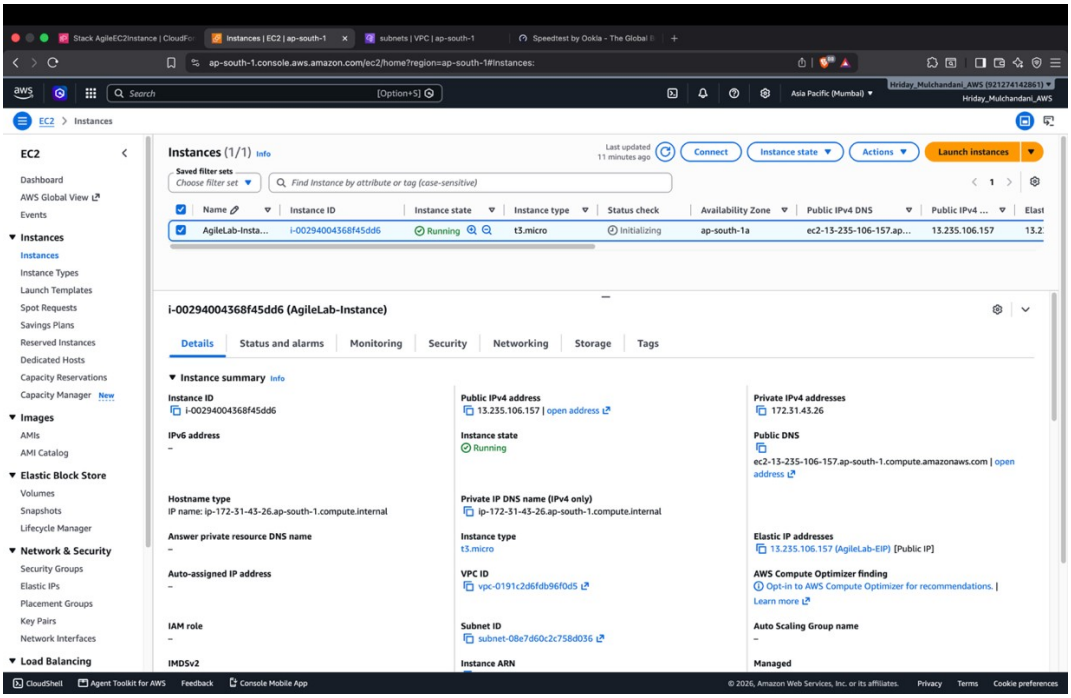
```
ElasticIP:
  Description: Elastic (Public) IP address
  Value: !Ref MyElasticIP
```

Output:






Instances (1/1)
i-00294004368f45dd6 (AgileLab-Instance)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary
Instance ID
i-00294004368f45dd6
Public IPv4 address
13.235.106.157 | open address
Private IPv4 addresses
172.31.43.26
IPv6 address
Instance state
Running
Public DNS
ec2-13-235-106-157.ap-south-1.compute.amazonaws.com | open address
Hostname type
IP name: ip-172-31-43-26.ap-south-1.compute.internal
Private IP DNS name (IPv4 only)
ip-172-31-43-26.ap-south-1.compute.internal
Answer private resource DNS name
Instance type
t3.micro
Elastic IP addresses
13.235.106.157 (AgileLab-EIP) [Public IP]


Instances (1/1)
i-00294004368f45dd6 (AgileLab-Instance)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary
Instance ID
i-00294004368f45dd6
Public IPv4 address
13.235.106.157 | open address
Private IPv4 addresses
172.31.43.26
IPv6 address
Instance state
Running
Public DNS
ec2-13-235-106-157.ap-south-1.compute.amazonaws.com | open address
Hostname type
IP name: ip-172-31-43-26.ap-south-1.compute.internal
Private IP DNS name (IPv4 only)
ip-172-31-43-26.ap-south-1.compute.internal
Answer private resource DNS name
Instance type
t3.micro
Elastic IP addresses
13.235.106.157 (AgileLab-EIP) [Public IP]
Auto-assigned IP address
VPC ID
vpc-0191c2d6fdb96f0d5
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
Learn more
IAM role
Subnet ID
subnet-08e7d60c2c758d036
Auto Scaling Group name
IMDSv2
Instance ARN
Managed

Lab 4

Aim:

JIRA Project Management tool

Theory:

1. Introduction to JIRA

JIRA is a project management and issue tracking tool developed by Atlassian. It is widely used in Agile software development to plan projects, assign tasks, track progress, and manage bugs. JIRA supports Scrum, Kanban, sprint planning, backlog management, and project reporting.

2. Features of JIRA

1. **Project Management:** Organizes and manages software projects.
2. **Issue Tracking:** Tracks tasks, bugs, and user stories.
3. **Scrum Board:** Displays task progress visually.
4. **Sprint Planning:** Helps plan work for a fixed development period.
5. **Backlog Management:** Maintains a list of pending work.
6. **Reports:** Provides burndown charts, velocity reports, and sprint reports.
7. **Collaboration:** Allows team members to assign tasks and monitor progress.

3. Important JIRA Concepts

- **Project:** A collection of related tasks and activities.
- **Epic:** A large feature divided into smaller user stories.
- **User Story:** A requirement describing what a user wants from the system.
- **Task:** A specific activity assigned to a team member.
- **Bug:** An error or defect in the software.
- **Sprint:** A fixed development period, usually 1 to 4 weeks.
- **Backlog:** A prioritized list of pending work.
- **Scrum Board:** A visual board used to track task progress.
- **Assignee:** The person responsible for completing a task.
- **Issue:** A general work item in JIRA, such as a task, bug, or story.

4. Practical Tasks Performed

Task 1: Create a JIRA Account

Created a JIRA account using an email address and logged into the platform.

Task 2: Create a Scrum Project

Created a team managed Scrum project with the following details:

Project Name: Student Management System

Project Key: SMS

Task 3: Create an Epic

Created an Epic named **Student Registration Module** to manage student registration functionality.

Task 4: Create User Stories

Created four user stories:

1. US1: Student registration into the portal.
2. US2: Admin verification of student details.
3. US3: Secure student login.
4. US4: Student profile updates.

Task 5: Create Tasks

Created and assigned four development tasks:

1. Design Registration Page
2. Create Database
3. Develop Login Page
4. Test Registration

All tasks were assigned to ourselves.

Task 6: Create Bugs

Created two bugs to track software defects:

1. BUG1: Login button not working.
2. BUG2: Registration page crashes on empty input.

Task 7: Create Sprint

Created **Sprint 1** with a duration of **2 weeks** and moved the user stories from the backlog into the sprint.

Task 8: Start Sprint

Started Sprint 1 to begin working on the selected user stories and tasks.

Task 9: Update Task Status

Updated task progress using the Scrum Board.

The workflow followed was:

To Do → In Progress → Testing → Done

Task 10: Generate Reports

Observed the following reports to monitor project progress:

- **Sprint Report:** Shows completed and incomplete sprint work.
- **Burndown Chart:** Shows remaining work during a sprint.
- **Velocity Report:** Shows the amount of work completed across sprints.
- **Backlog View:** Displays pending tasks and user stories.

5. Backlog vs Sprint

Backlog:

1. Contains all pending project work.
2. Includes Epics, User Stories, Tasks, and Bugs.
3. Managed and prioritized by the Product Owner.
4. Continuously updated throughout the project.

Sprint:

1. Contains selected work from the backlog.
2. Has a fixed duration, usually 1 to 4 weeks.
3. Executed by the Scrum Team.
4. Focuses on completing planned work.

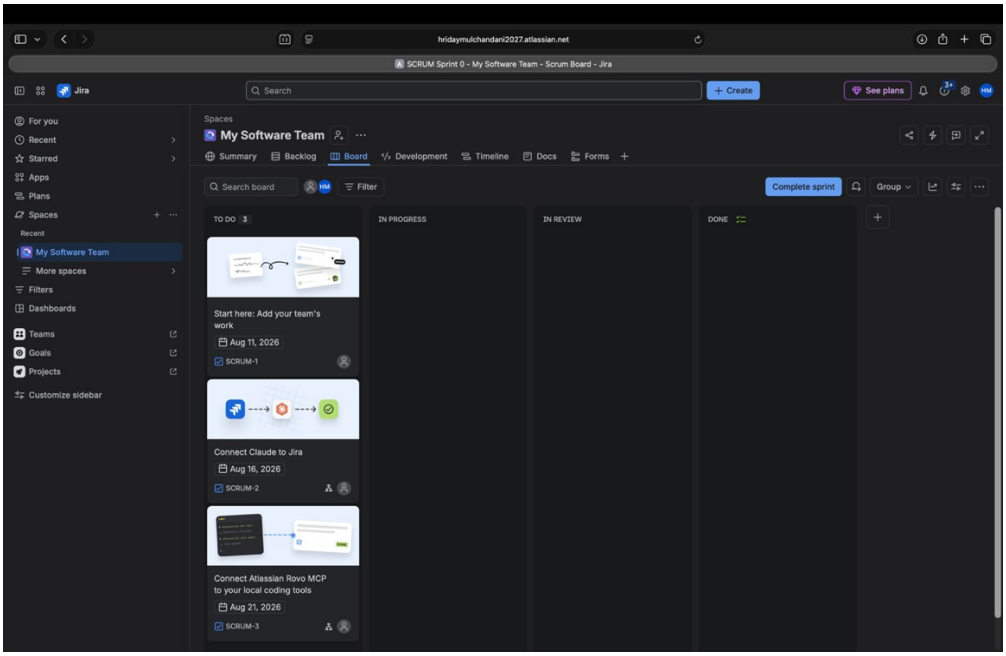
6. JIRA Workflow

The general workflow followed in JIRA is:

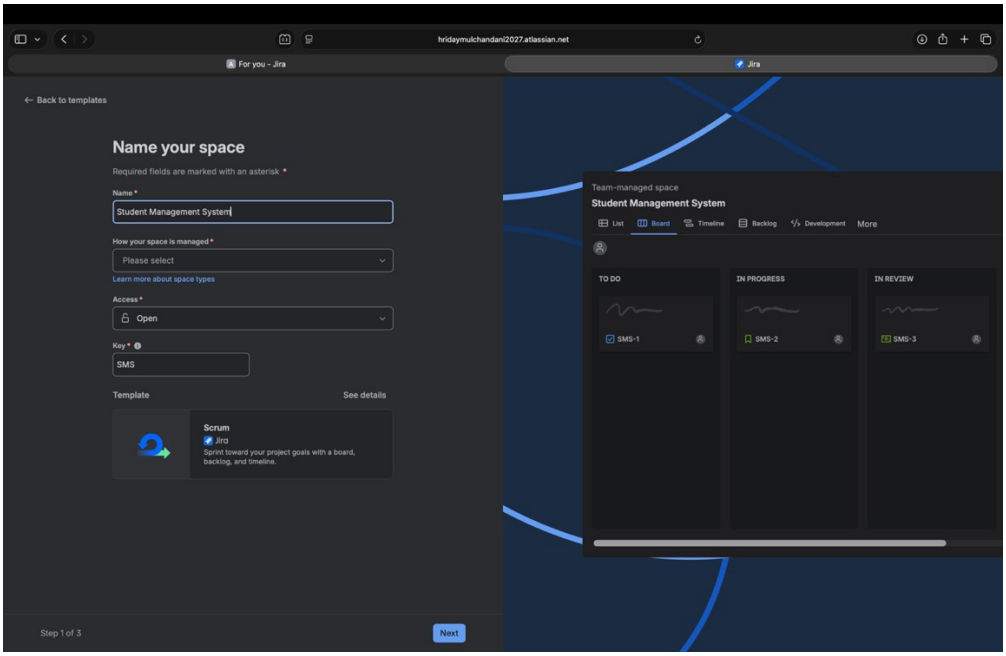
1. Identify project requirements.
2. Create Epics and User Stories.
3. Create Tasks and Bugs.
4. Add work items to the Product Backlog.
5. Plan and start a Sprint.
6. Track progress using the Scrum Board.
7. Review completed sprint work.
8. Generate reports to monitor progress.

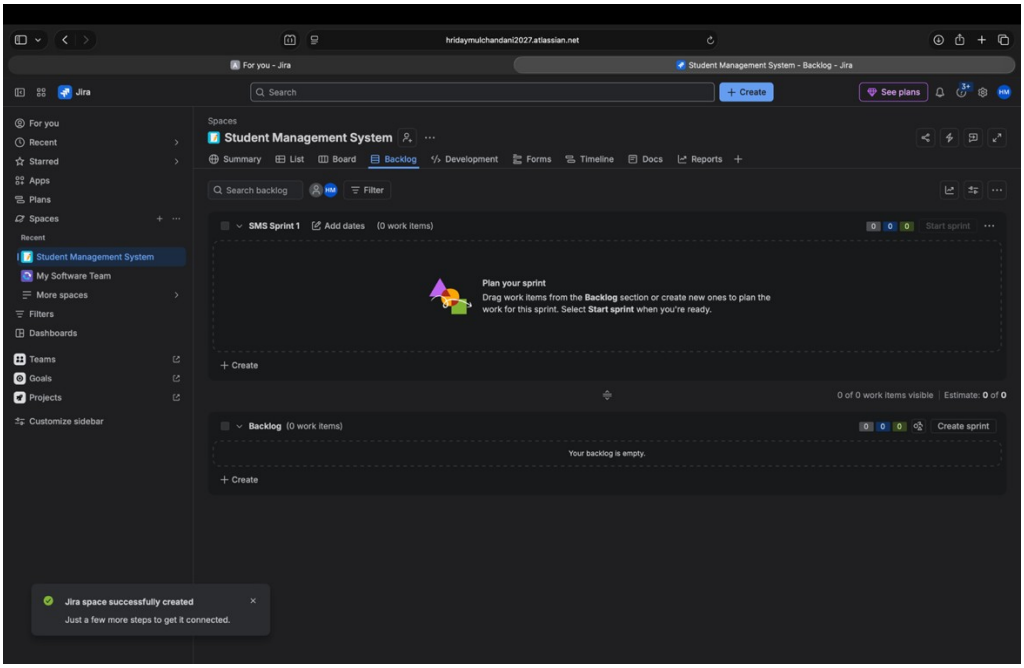
Output:

Task – 1

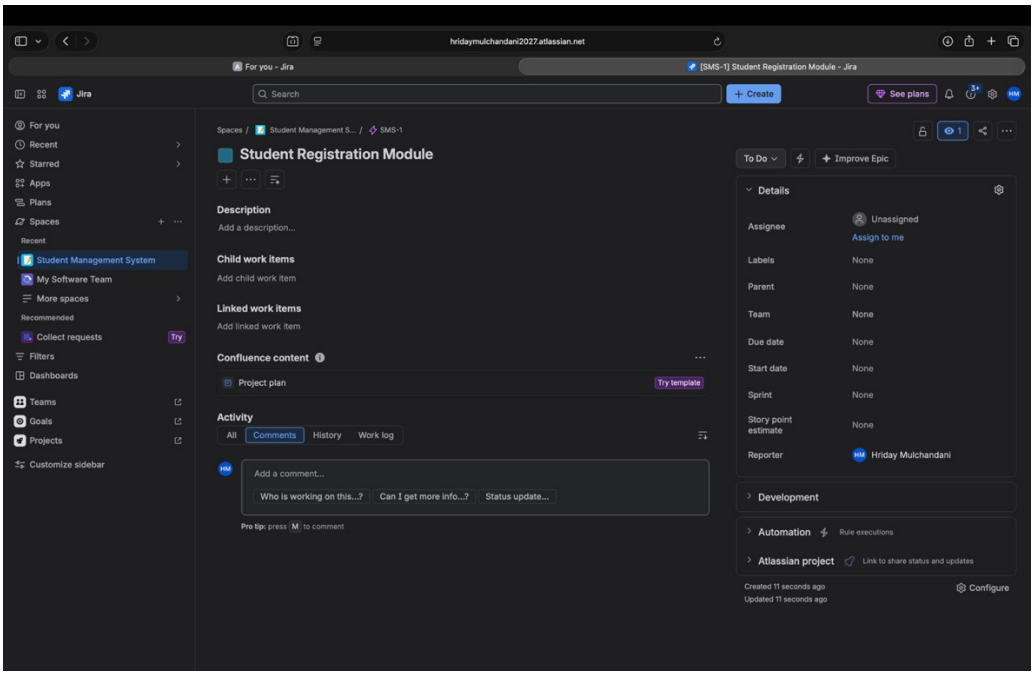

hridaymulchandani2027.atlassian.net
SCRUM Sprint 0 - My Software Team - Scrum Board - Jira
Jira
Search
Create
See plans
For you
Recent
Starred
Apps
Plans
Spaces
Recent
My Software Team
More spaces
Filters
Dashboards
Teams
Goals
Projects
Customize sidebar
Spaces
My Software Team
Summary
Backlog
Board
Development
Timeline
Docs
Forms
Search board
Filter
Complete sprint
Group
TO DO 3
IN PROGRESS
IN REVIEW
DONE
Start here: Add your team's work
Aug 11, 2026
SCRUM-1
Connect Claude to Jira
Aug 16, 2026
SCRUM-2
Connect Atlassian Rovo MCP to your local coding tools
Aug 21, 2026
SCRUM-3

Task – 2

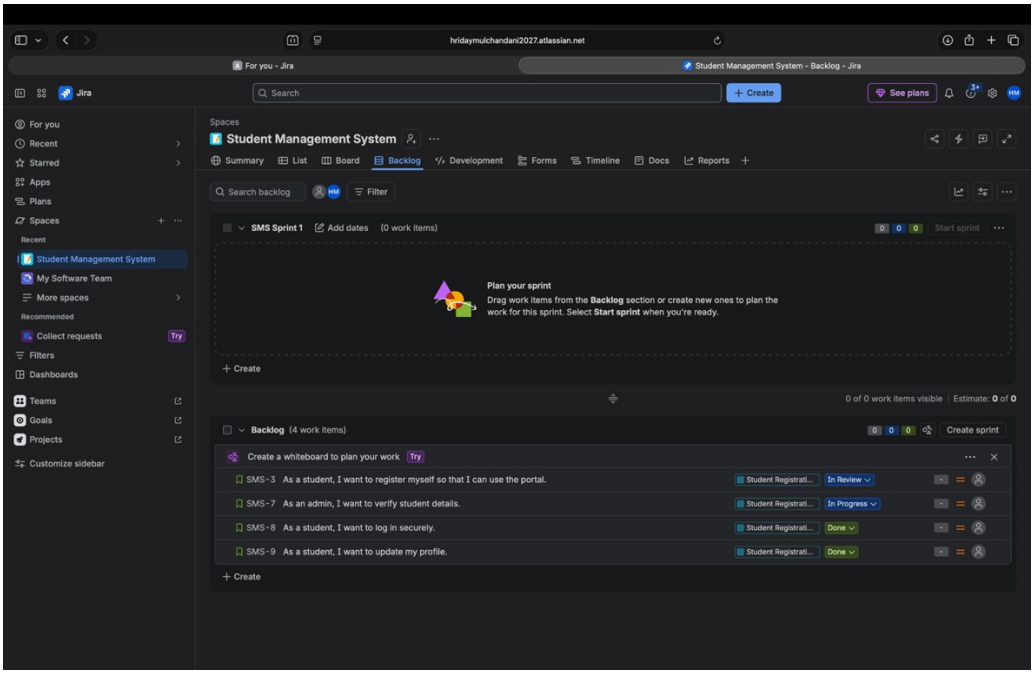

hridaymulchandani2027.atlassian.net
For you - Jira
Jira
Back to templates
Name your space
Required fields are marked with an asterisk *
Name *
Student Management System
How your space is managed *
Please select
Learn more about space types
Access *
Open
Key *
SMS
Template
See details
Scrum
Jira
Sprint toward your project goals with a board, backlog, and timeline.
Step 1 of 3
Next
Team-managed space
Student Management System
List
Board
Timeline
Backlog
Development
More
TO DO
IN PROGRESS
IN REVIEW
SMS-1
SMS-2
SMS-3


hridaymulchandani2027.atlassian.net
For you - Jira
Student Management System - Backlog - Jira
Jira
Search
Create
See plans
Spaces
Student Management System
Summary
List
Board
Backlog
Development
Forms
Timeline
Docs
Reports
Search backlog
Filter
SMS Sprint 1
Add dates
(0 work items)
Start sprint
Plan your sprint
Drag work items from the Backlog section or create new ones to plan the work for this sprint. Select Start sprint when you're ready.
Create
0 of 0 work items visible | Estimate: 0 of 0
Backlog (0 work items)
Create sprint
Your backlog is empty.
Create
Jira space successfully created
Just a few more steps to get it connected.

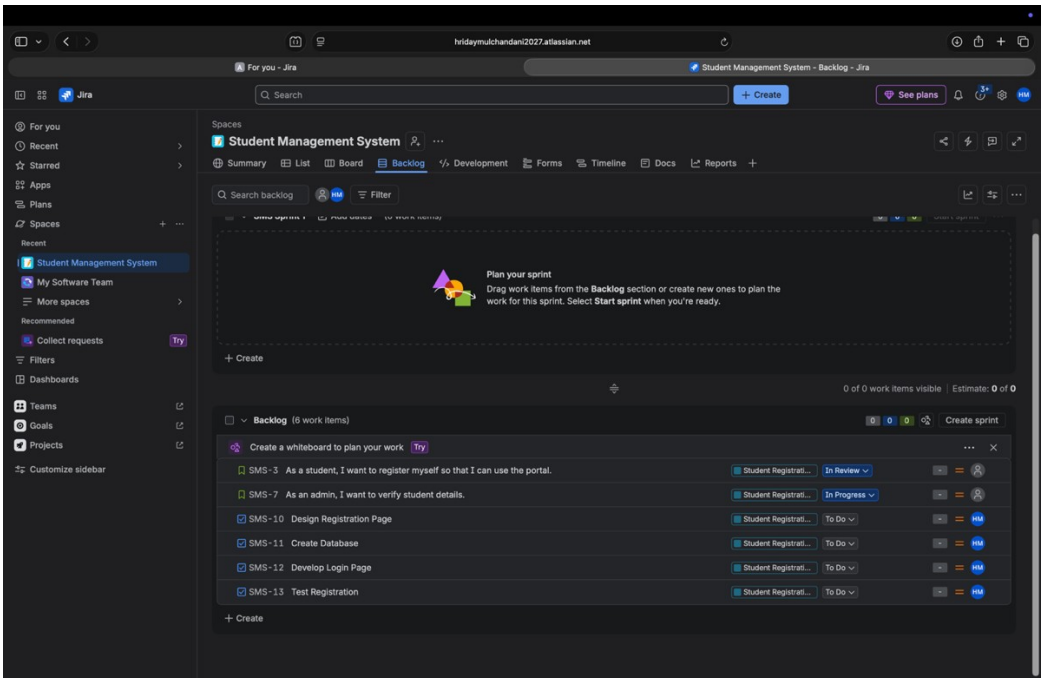
Task – 3


hridaymulchandani2027.atlassian.net
For you - Jira
[SMS-1] Student Registration Module - Jira
Jira
Search
Create
See plans
Spaces / Student Management S... / SMS-1
Student Registration Module
To Do
Improve Epic
Description
Add a description...
Child work items
Add child work item
Linked work items
Add linked work item
Confluence content
Project plan
Try template
Activity
All
Comments
History
Work log
Add a comment...
Who is working on this...?
Can I get more info...?
Status update...
Pro tip: press M to comment
Details
Assignee
Unassigned
Assign to me
Labels
None
Parent
None
Team
None
Due date
None
Start date
None
Sprint
None
Story point estimate
None
Reporter
Hriday Mulchandani
Development
Automation
Rule executions
Atlassian project
Link to share status and updates
Created 11 seconds ago
Updated 11 seconds ago
Configure

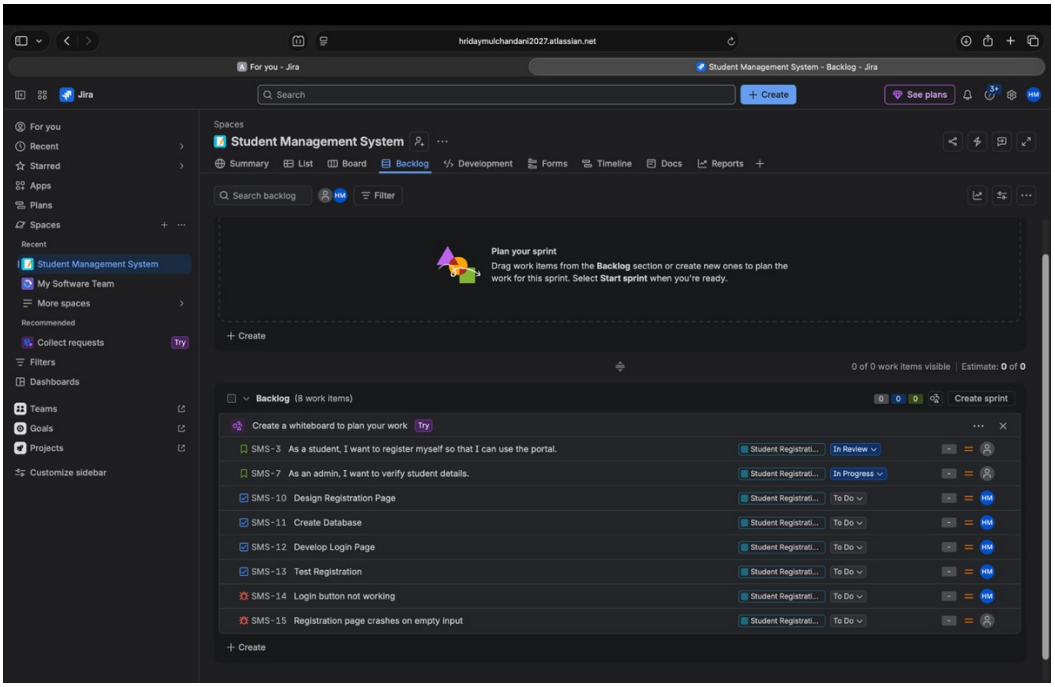
Task – 4


hridaymulchandani2027.atlassian.net
For you - Jira
Student Management System - Backlog - Jira
Jira
Search
Create
See plans
Spaces
Student Management System
Summary
List
Board
Backlog
Development
Forms
Timeline
Docs
Reports
Search backlog
Filter
SMS Sprint 1
Add dates
(0 work items)
Start sprint
Plan your sprint
Drag work items from the Backlog section or create new ones to plan the work for this sprint. Select Start sprint when you're ready.
Create
0 of 0 work items visible
Estimate: 0 of 0
Backlog (4 work items)
Create sprint
Create a whiteboard to plan your work
Try
SMS-3 As a student, I want to register myself so that I can use the portal.
SMS-7 As an admin, I want to verify student details.
SMS-8 As a student, I want to log in securely.
SMS-9 As a student, I want to update my profile.
Student Registrati...
In Review
In Progress
Done
For you
Recent
Starred
Apps
Plans
Spaces
Recent
Student Management System
My Software Team
More spaces
Recommended
Collect requests
Filters
Dashboards
Teams
Goals
Projects
Customize sidebar

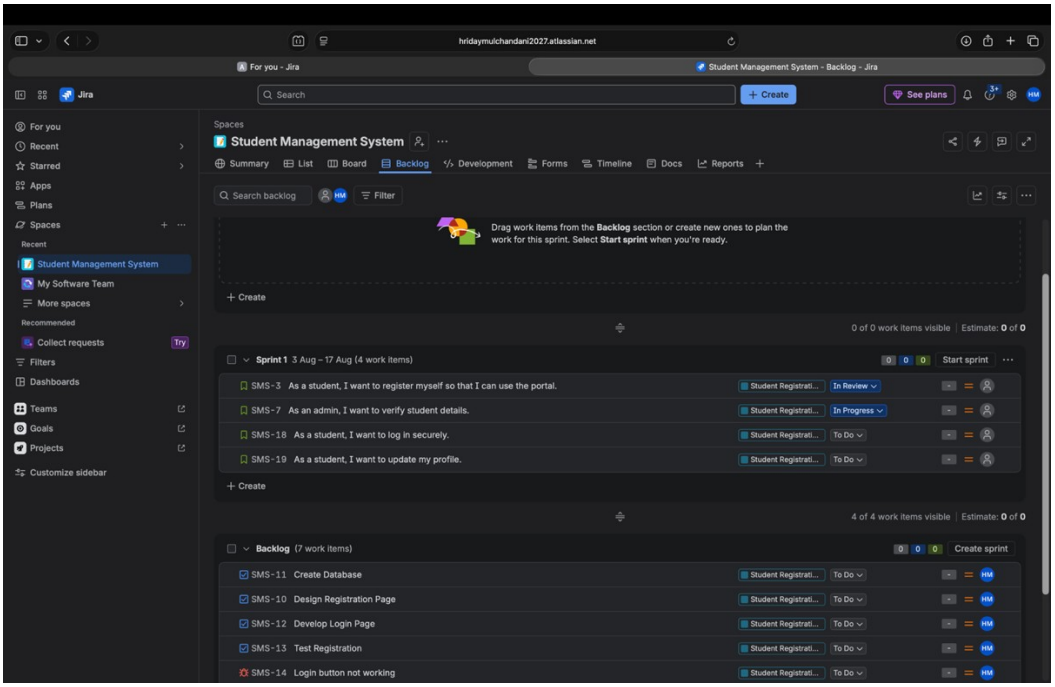
Task – 5


hridaymulchandani2027.atlassian.net
For you - Jira
Student Management System - Backlog - Jira
Jira
Search
Create
See plans
Spaces
Student Management System
Summary
List
Board
Backlog
Development
Forms
Timeline
Docs
Reports
Search backlog
Filter
Plan your sprint
Drag work items from the Backlog section or create new ones to plan the work for this sprint. Select Start sprint when you're ready.
Create
0 of 0 work items visible
Estimate: 0 of 0
Backlog (6 work items)
Create sprint
Create a whiteboard to plan your work
Try
SMS-3 As a student, I want to register myself so that I can use the portal.
SMS-7 As an admin, I want to verify student details.
SMS-10 Design Registration Page
SMS-11 Create Database
SMS-12 Develop Login Page
SMS-13 Test Registration
Student Registrati...
In Review
In Progress
To Do
For you
Recent
Starred
Apps
Plans
Spaces
Recent
Student Management System
My Software Team
More spaces
Recommended
Collect requests
Filters
Dashboards
Teams
Goals
Projects
Customize sidebar

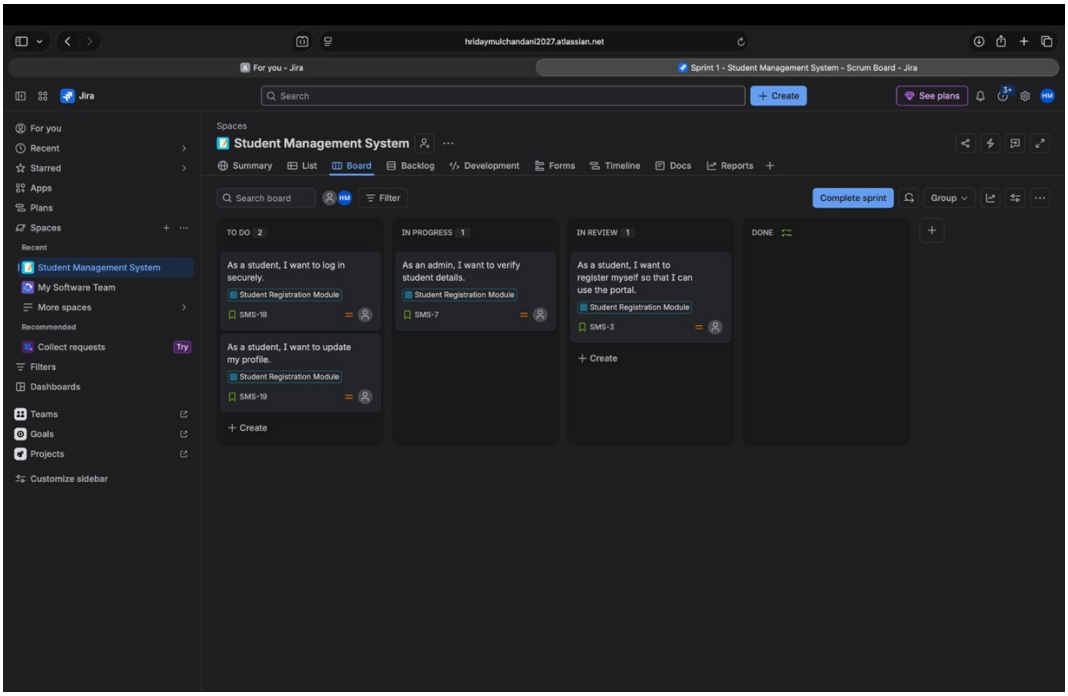
Task – 6


hridaymulchandani2027.atlassian.net
For you - Jira
Student Management System - Backlog - Jira
Jira
Search
Create
See plans
For you
Recent
Starred
Apps
Plans
Spaces
Recent
Student Management System
My Software Team
More spaces
Recommended
Collect requests
Try
Filters
Dashboards
Teams
Goals
Projects
Customize sidebar
Spaces
Student Management System
Summary
List
Board
Backlog
Development
Forms
Timeline
Docs
Reports
Search backlog
Filter
Plan your sprint
Drag work items from the Backlog section or create new ones to plan the work for this sprint. Select Start sprint when you're ready.
Create
0 of 0 work items visible
Estimate: 0 of 0
Backlog (8 work items)
Create sprint
Create a whiteboard to plan your work
Try
SMS-3 As a student, I want to register myself so that I can use the portal.
SMS-7 As an admin, I want to verify student details.
SMS-10 Design Registration Page
SMS-11 Create Database
SMS-12 Develop Login Page
SMS-13 Test Registration
SMS-14 Login button not working
SMS-15 Registration page crashes on empty input
Student Registrati...
In Review
In Progress
To Do
Create

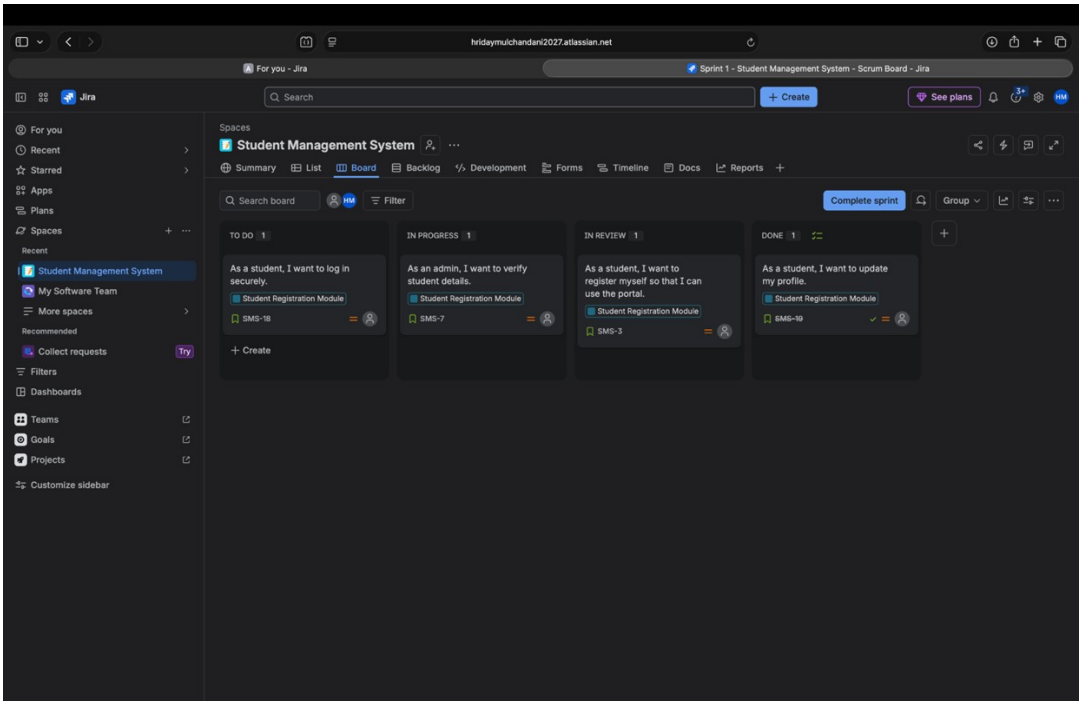
Task – 7


hridaymulchandani2027.atlassian.net
For you - Jira
Student Management System - Backlog - Jira
Jira
Search
Create
See plans
For you
Recent
Starred
Apps
Plans
Spaces
Recent
Student Management System
My Software Team
More spaces
Recommended
Collect requests
Try
Filters
Dashboards
Teams
Goals
Projects
Customize sidebar
Spaces
Student Management System
Summary
List
Board
Backlog
Development
Forms
Timeline
Docs
Reports
Search backlog
Filter
Drag work items from the Backlog section or create new ones to plan the work for this sprint. Select Start sprint when you're ready.
Create
0 of 0 work items visible
Estimate: 0 of 0
Sprint 1 3 Aug – 17 Aug (4 work items)
Start sprint
SMS-3 As a student, I want to register myself so that I can use the portal.
SMS-7 As an admin, I want to verify student details.
SMS-18 As a student, I want to log in securely.
SMS-19 As a student, I want to update my profile.
Student Registrati...
In Review
In Progress
To Do
Create
4 of 4 work items visible
Estimate: 0 of 0
Backlog (7 work items)
Create sprint
SMS-11 Create Database
SMS-10 Design Registration Page
SMS-12 Develop Login Page
SMS-13 Test Registration
SMS-14 Login button not working

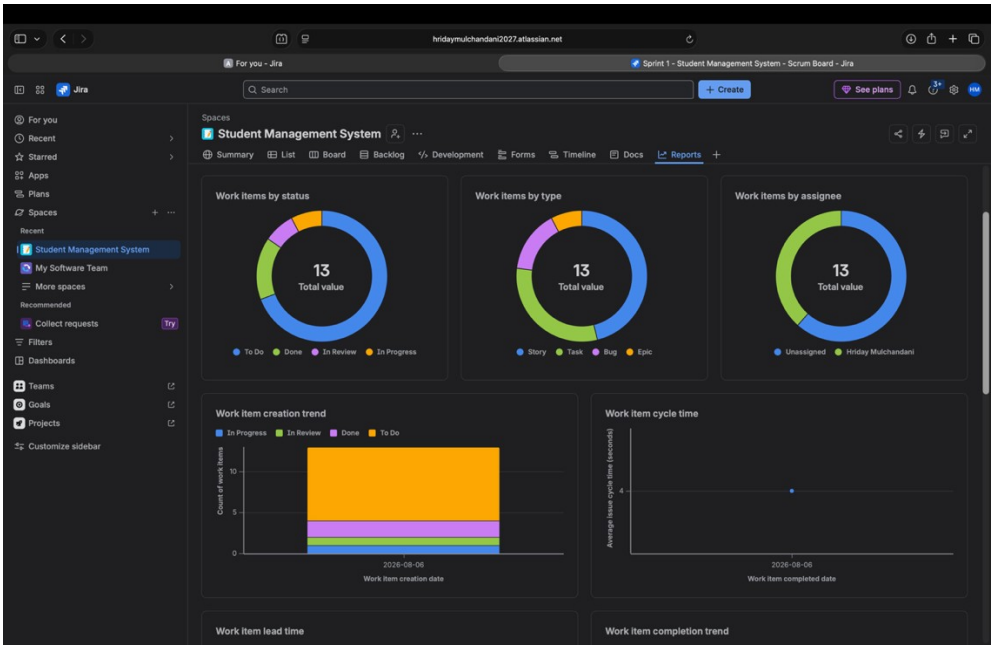
Task – 8

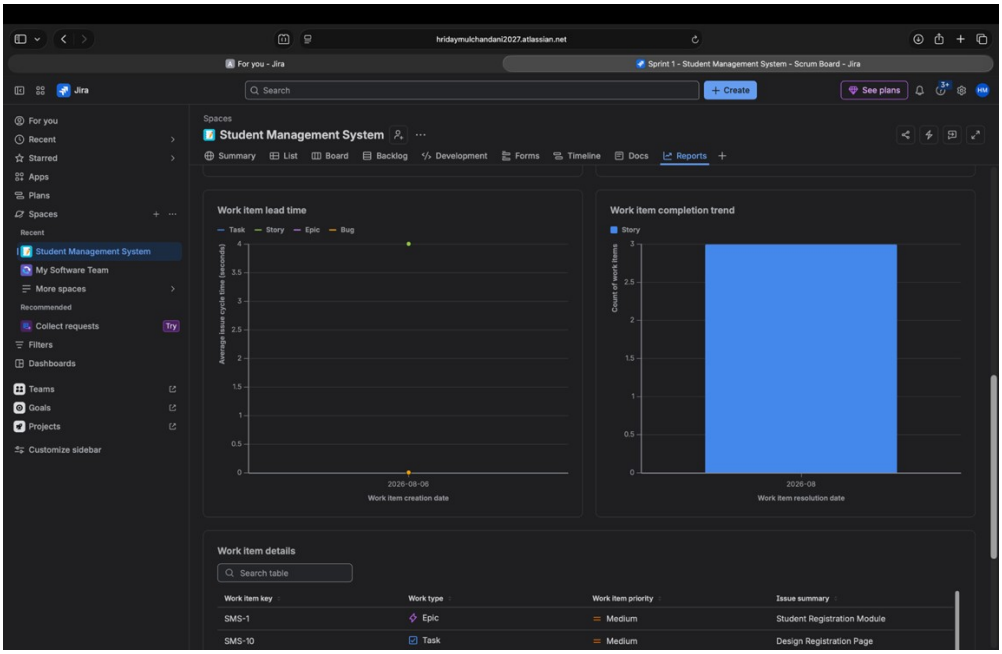

hridaymulchandani2027.atlassian.net
For you - Jira
Sprint 1 - Student Management System - Scrum Board - Jira
Jira
Search
Create
See plans
For you
Recent
Starred
Apps
Plans
Spaces
Recent
Student Management System
My Software Team
More spaces
Recommended
Collect requests
Try
Filters
Dashboards
Teams
Goals
Projects
Customize sidebar
Spaces
Student Management System
Summary
List
Board
Backlog
Development
Forms
Timeline
Docs
Reports
Search board
Filter
Complete sprint
Group
TO DO 2
IN PROGRESS 1
IN REVIEW 1
DONE
As a student, I want to log in securely.
Student Registration Module
SMS-18
As a student, I want to update my profile.
Student Registration Module
SMS-19
Create
As an admin, I want to verify student details.
Student Registration Module
SMS-7
As a student, I want to register myself so that I can use the portal.
Student Registration Module
SMS-3
Create

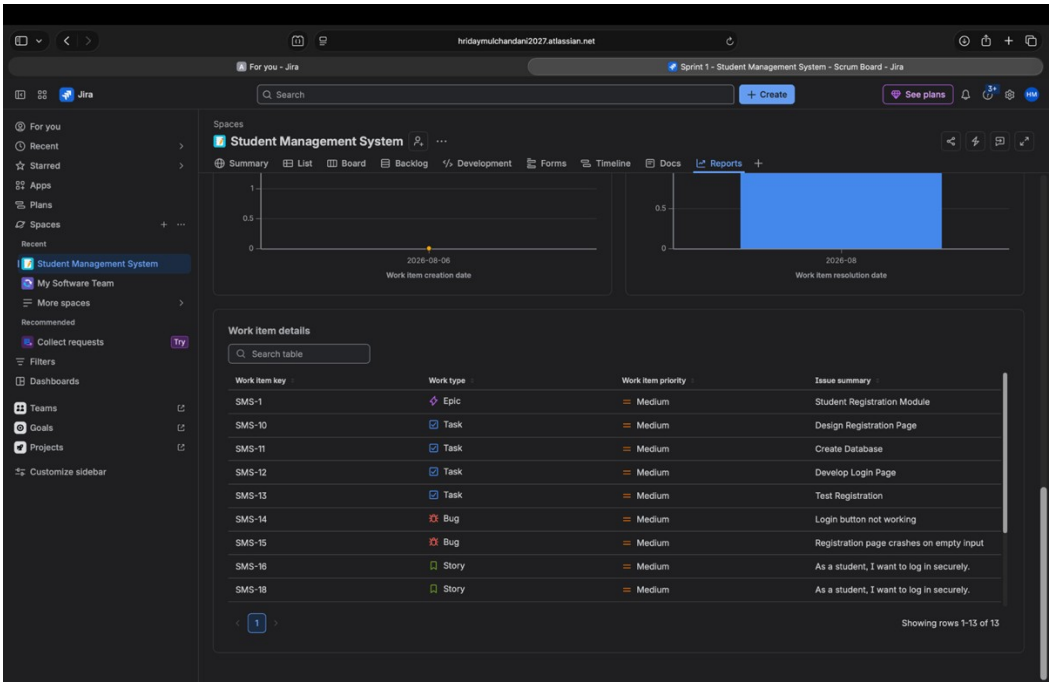
Task – 9

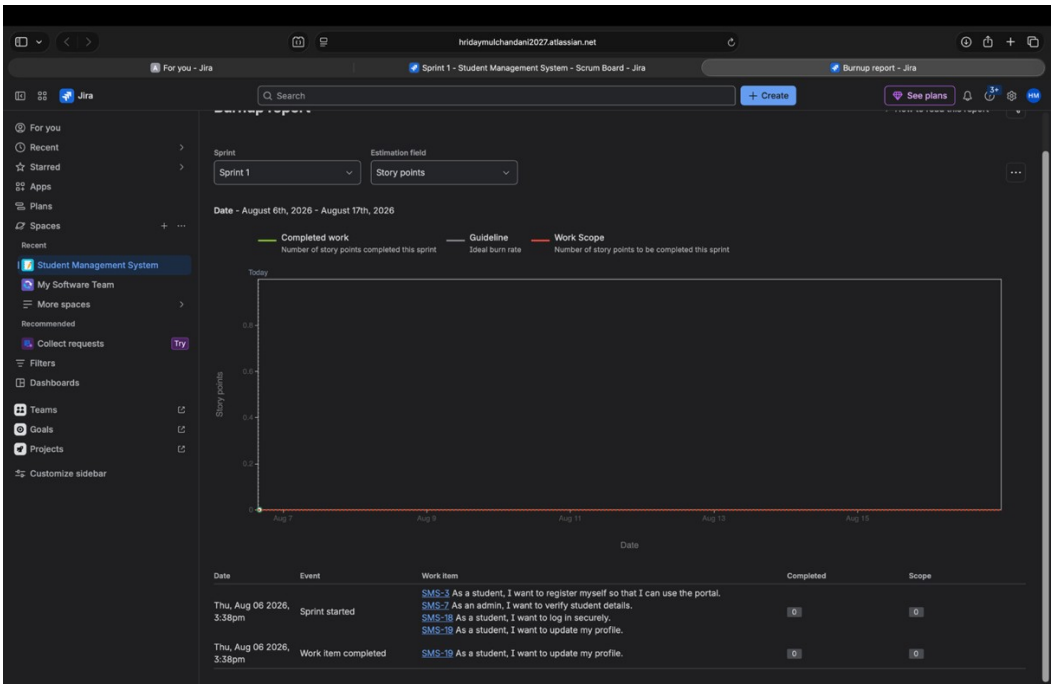

hridaymulchandani2027.atlassian.net
For you - Jira
Sprint 1 - Student Management System - Scrum Board - Jira
Jira
Search
Create
See plans
For you
Recent
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Student Management System
My Software Team
More spaces
Recommended
Collect requests
Try
Filters
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Customize sidebar
Spaces
Student Management System
Summary
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Timeline
Docs
Reports
Search board
Filter
Complete sprint
Group
TO DO 1
IN PROGRESS 1
IN REVIEW 1
DONE 1
As a student, I want to log in securely.
Student Registration Module
SMS-18
Create
As an admin, I want to verify student details.
Student Registration Module
SMS-7
As a student, I want to register myself so that I can use the portal.
Student Registration Module
SMS-3
As a student, I want to update my profile.
Student Registration Module
SMS-19

Task – 10


hridaymulchandani2027.atlassian.net
For you - Jira
Sprint 1 - Student Management System - Scrum Board - Jira
Jira
Search
Create
See plans
Spaces
Student Management System
Summary
List
Board
Backlog
Development
Forms
Timeline
Docs
Reports
Work items by status
13
Total value
To Do
Done
In Review
In Progress
Work items by type
13
Total value
Story
Task
Bug
Epic
Work items by assignee
13
Total value
Unassigned
Hriday Mulchandani
Work item creation trend
In Progress
In Review
Done
To Do
Count of work items
2026-08-06
Work item creation date
Work item cycle time
Average issue cycle time (seconds)
2026-08-06
Work item completed date
Work item lead time
Work item completion trend
For you
Recent
Starred
Apps
Plans
Spaces
Recent
Student Management System
My Software Team
More spaces
Recommended
Collect requests
Try
Filters
Dashboards
Teams
Goals
Projects
Customize sidebar


hridaymulchandani2027.atlassian.net
For you - Jira
Sprint 1 - Student Management System - Scrum Board - Jira
Jira
Search
Create
See plans
Spaces
Student Management System
Summary
List
Board
Backlog
Development
Forms
Timeline
Docs
Reports
Work item lead time
Task
Story
Epic
Bug
Average issue cycle time (seconds)
2026-08-06
Work item creation date
Work item completion trend
Story
Count of work items
2026-08
Work item resolution date
Work item details
Search table
Work item key
Work type
Work item priority
Issue summary
SMS-1
Epic
Medium
Student Registration Module
SMS-10
Task
Medium
Design Registration Page
For you
Recent
Starred
Apps
Plans
Spaces
Recent
Student Management System
My Software Team
More spaces
Recommended
Collect requests
Try
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Projects
Customize sidebar


hridaymulchandani2027.atlassian.net
For you - Jira
Sprint 1 - Student Management System - Scrum Board - Jira
Jira
Search
Create
See plans
For you
Recent
Starred
Apps
Plans
Spaces
Recent
Student Management System
My Software Team
More spaces
Recommended
Collect requests
Try
Filters
Dashboards
Teams
Goals
Projects
Customize sidebar
Spaces
Student Management System
Summary
List
Board
Backlog
Development
Forms
Timeline
Docs
Reports
2026-08-06
Work item creation date
2026-08
Work item resolution date
Work item details
Search table
Work item key
Work type
Work item priority
Issue summary
SMS-1 Epic Medium Student Registration Module
SMS-10 Task Medium Design Registration Page
SMS-11 Task Medium Create Database
SMS-12 Task Medium Develop Login Page
SMS-13 Task Medium Test Registration
SMS-14 Bug Medium Login button not working
SMS-15 Bug Medium Registration page crashes on empty input
SMS-16 Story Medium As a student, I want to log in securely.
SMS-18 Story Medium As a student, I want to log in securely.
Showing rows 1-13 of 13


hridaymulchandani2027.atlassian.net
For you - Jira
Sprint 1 - Student Management System - Scrum Board - Jira
Burnup report - Jira
Jira
Search
Create
See plans
Sprint
Sprint 1
Estimation field
Story points
Date - August 6th, 2026 - August 17th, 2026
Completed work
Number of story points completed this sprint
Guideline
Ideal burn rate
Work Scope
Number of story points to be completed this sprint
Today
Story points
Aug 7
Aug 9
Aug 11
Aug 13
Aug 15
Date
Date
Event
Work item
Completed
Scope
Thu, Aug 06 2026, 3:38pm
Sprint started
SMS-3 As a student, I want to register myself so that I can use the portal.
SMS-7 As an admin, I want to verify student details.
SMS-18 As a student, I want to log in securely.
SMS-19 As a student, I want to update my profile.
0
0
Thu, Aug 06 2026, 3:38pm
Work item completed
SMS-19 As a student, I want to update my profile.
0
0

Lab 5 and 6

Aim:

GitHub Actions and GitHub Actions Workflows

Theory:

1. Introduction to GitHub Actions

GitHub Actions is an automation tool provided by GitHub that allows developers to automate software development tasks such as building, testing, and deploying applications. It is mainly used to implement Continuous Integration and Continuous Deployment (CI/CD) directly within GitHub repositories.

2. Features of GitHub Actions

1. **Automation:** Automatically executes tasks when specific events occur.
2. **CI/CD:** Automates building, testing, and deployment of applications.
3. **Event Based Execution:** Triggers workflows when code is pushed or pull requests are created.
4. **Multiple Platforms:** Supports Linux, Windows, and macOS.
5. **Integration:** Works with GitHub repositories and external services.
6. **Monitoring:** Provides logs and execution status of workflows.

3. Introduction to GitHub Actions Workflows

A GitHub Actions Workflow is an automated process consisting of one or more jobs. It is defined using a YAML file stored inside the .github/workflows directory of a repository.

Workflows are triggered by events such as pushing code, creating pull requests, or manual execution.

4. Important Components of GitHub Actions

- **Workflow:** An automated process containing one or more jobs.
- **Event:** An activity that triggers a workflow, such as push or pull request.
- **Job:** A collection of steps executed on the same runner.
- **Step:** An individual task within a job, such as installing dependencies or running tests.
- **Action:** A reusable component that performs a specific task within a workflow.
- **Runner:** A machine that executes workflow jobs. It can run on Linux, Windows, or macOS.
- **YAML File:** A configuration file used to define workflows, jobs, steps, and triggers.
- **Secrets:** Securely stored sensitive values, such as API keys and authentication tokens, used by workflows.

5. Continuous Integration and Continuous Deployment

Continuous Integration (CI):

CI is the practice of frequently merging code changes into a shared repository. Automated builds and tests are performed to detect errors early.

Continuous Deployment (CD):

Continuous Deployment automatically releases tested code to production without manual approval.

Advantages of CI/CD:

1. Reduces manual work.
2. Detects errors early.
3. Improves software quality.
4. Speeds up application delivery.
5. Provides consistent build and deployment processes.

6. GitHub Actions Workflow Triggers

Workflows can execute automatically or manually based on specific events.

- **Push:** Runs when code is pushed to a repository.
- **Pull Request:** Runs when pull request activities occur.
- **Workflow Dispatch:** Allows manual execution of a workflow.
- **Schedule:** Executes workflows at specified times.
- **Release:** Runs when specified release events occur.

7. GitHub Actions Workflow Execution

The general workflow execution process is:

1. Developer pushes code to a GitHub repository.
2. GitHub detects the configured event.
3. The workflow is automatically triggered.
4. A runner is assigned to execute the job.
5. Dependencies are installed and the application is built.
6. Automated tests are executed.
7. The application may be deployed if configured.
8. Execution results and logs are displayed in GitHub Actions.

8. Practical Tasks

Task 1: Create a GitHub Repository

Create a new GitHub repository to store the application code and workflow files.

Task 2: Create a Workflow File

Create a YAML workflow file inside the .github/workflows directory to define automation instructions.

Task 3: Configure Workflow Triggers

Configure events such as push or pull request to automatically execute the workflow.

Task 4: Define Jobs and Steps

Specify jobs containing steps such as checking out code, installing dependencies, building the application, and running tests.

Task 5: Execute the Workflow

Push the workflow file to GitHub to trigger automatic execution.

Task 6: Monitor Workflow Execution

Open the Actions tab in GitHub to view running, successful, or failed workflows.

Task 7: Verify Workflow Results

Check execution logs and confirm whether the build and testing processes completed successfully.

9. Advantages of GitHub Actions

1. Provides built in automation within GitHub.
2. Reduces repetitive development tasks.
3. Supports automated testing and deployment.
4. Helps identify errors before deployment.
5. Supports reusable workflows and actions.
6. Improves development speed and collaboration.

Source Code:

```yaml
name: Calculator CI

on:
  push:
    branches: [main]
  pull_request:
    branches: [main]

jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout code
        uses: actions/checkout@v4

      - name: Set up JDK
        uses: actions/setup-java@v4
        with:
          distribution: temurin
          java-version: '17'

      - name: Compile
        run: javac Calculator.java

      - name: Upload compiled class
        uses: actions/upload-artifact@v4
        with:
          name: calculator-class
          path: Calculator.class
          retention-days: 1

  test:
    needs: build
    runs-on: ubuntu-latest
    strategy:
      fail-fast: false
      matrix:
        case:
          - { name: "addition",        a: "3",  op: "+", b: "4", expected: "7",   expect_failure: false }
          - { name: "negative-operand", a: "-5", op: "+", b: "3", expected: "-2",  expect_failure: false }
          - { name: "subtraction",     a: "10", op: "-", b: "4", expected: "6",   expect_failure: false }
          - { name: "multiplication",  a: "6",  op: "*", b: "7", expected: "42",  expect_failure: false }
          - { name: "division",        a: "10", op: "/", b: "2", expected: "5",   expect_failure: false }
          - { name: "division-decimal", a: "7", op: "/", b: "2", expected: "3.5", expect_failure: false }
          - { name: "divide-by-zero", a: "5",  op: "/", b: "0", expected: "",    expect_failure: true }
```

```
      - { name: "invalid-operator", a: "5", op: "%", b: "2", expected: "",   expect_failure: true }
      - { name: "non-numeric-operand", a: "abc", op: "+", b: "2", expected: "", expect_failure: true }
  steps:
    - name: Set up JDK
      uses: actions/setup-java@v4
      with:
        distribution: temurin
        java-version: '17'
    - name: Download compiled class
      uses: actions/download-artifact@v4
      with:
        name: calculator-class

    - name: "Run case: ${{ matrix.case.name }}"
      env:
        A: ${{ matrix.case.a }}
        OP: ${{ matrix.case.op }}
        B: ${{ matrix.case.b }}
        EXPECTED: ${{ matrix.case.expected }}
        EXPECT_FAILURE: ${{ matrix.case.expect_failure }}
      run: |
        set +e
        actual=$(java Calculator "$A" "$OP" "$B" 2>&1)
        exit_code=$?
        set -e

        if [ "$EXPECT_FAILURE" = "true" ]; then
          if [ "$exit_code" -eq 0 ]; then
            echo "FAIL: expected non-zero exit code, got 0 (output: $actual)"
            exit 1
          fi
          echo "PASS: correctly failed with: $actual"
        else
          if [ "$exit_code" -ne 0 ]; then
            echo "FAIL: expected success, got exit code $exit_code (output: $actual)"
            exit 1
          fi
          if [ "$actual" != "$EXPECTED" ]; then
            echo "FAIL: expected '$EXPECTED', got '$actual'"
            exit 1
          fi
          echo "PASS: $A $OP $B = $actual"
        fi

  missing-args:
    needs: build
    runs-on: ubuntu-latest
    steps:
```

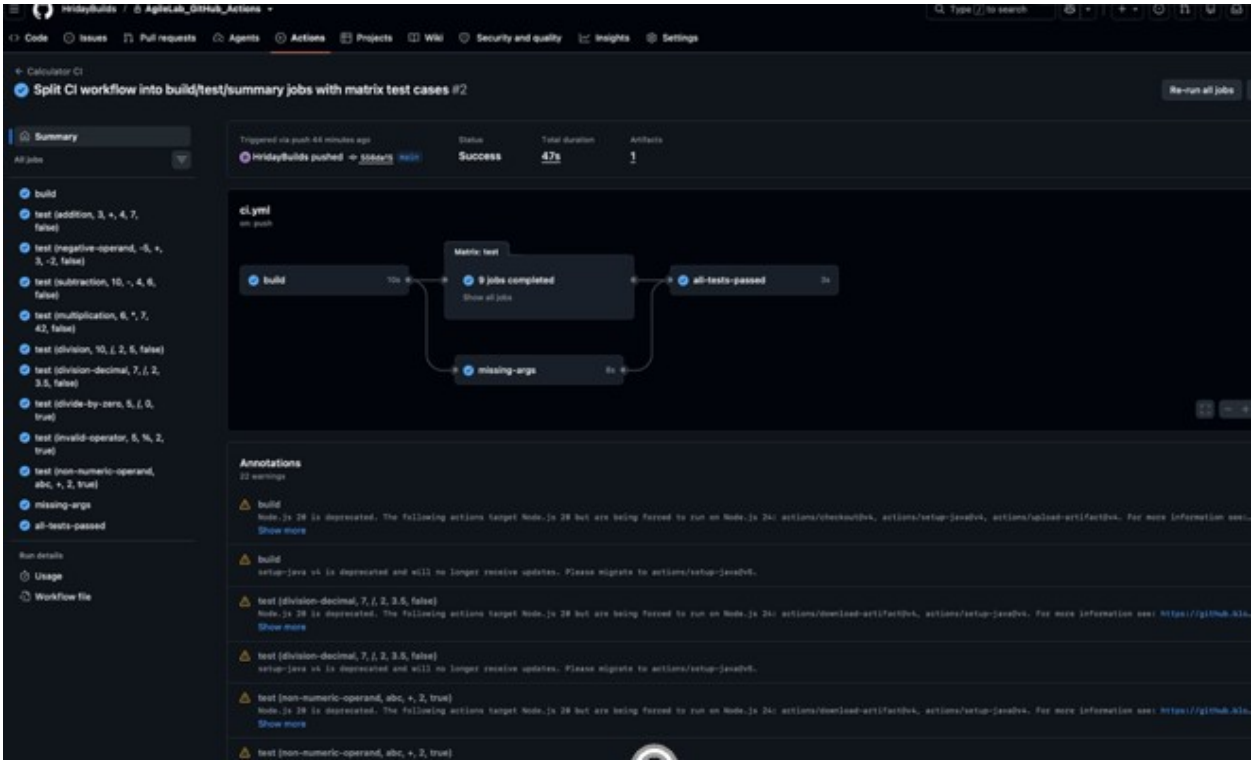
```yaml
    - name: Set up JDK
      uses: actions/setup-java@v4
      with:
        distribution: temurin
        java-version: '17'

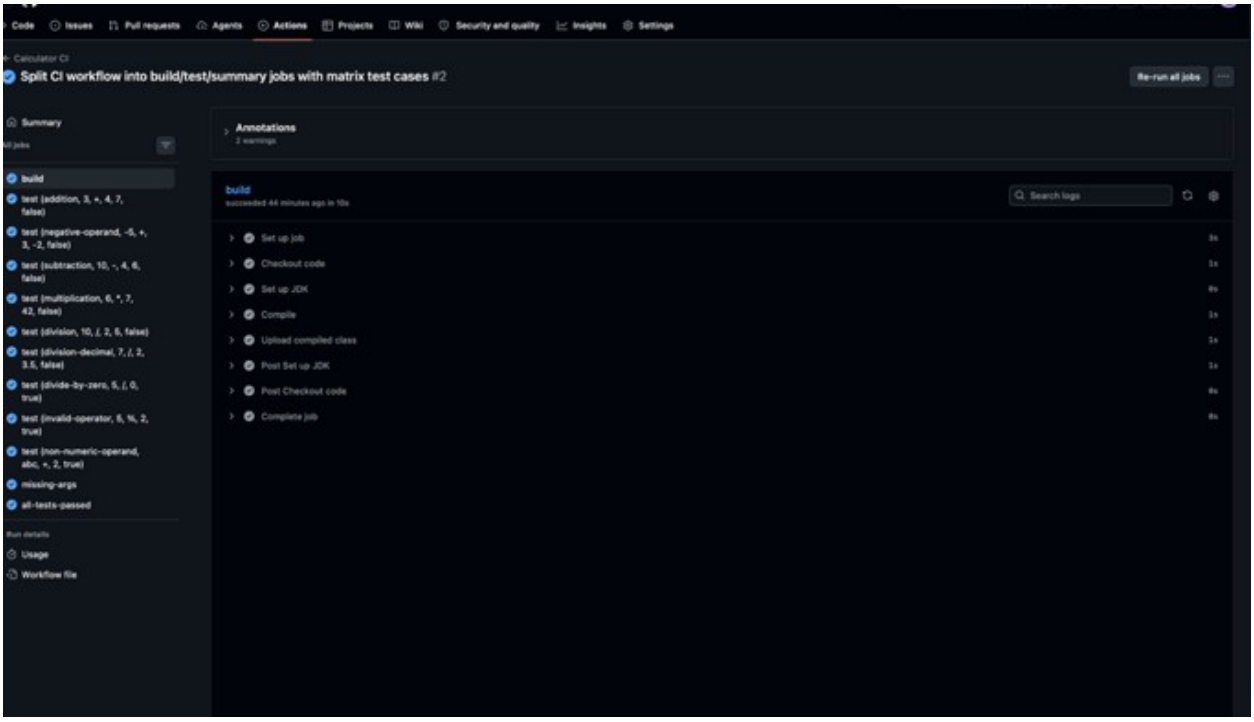
    - name: Download compiled class
      uses: actions/download-artifact@v4
      with:
        name: calculator-class

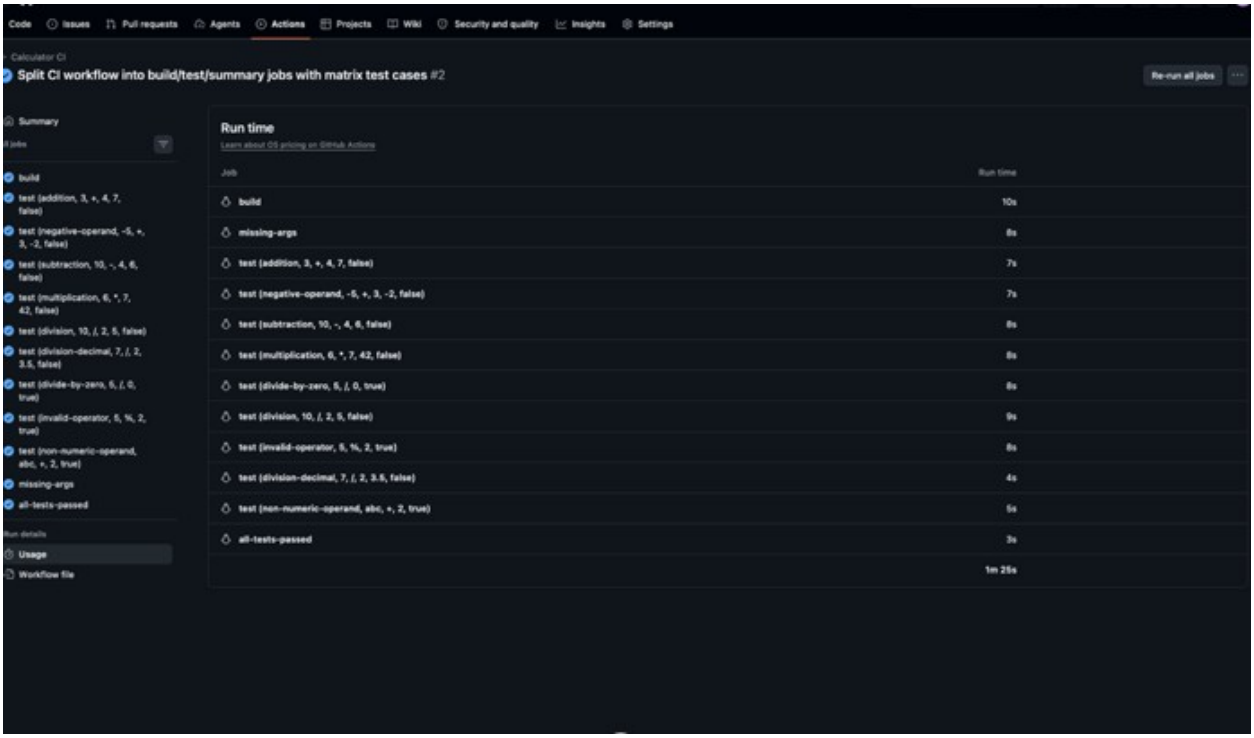
    - name: Missing arguments should fail
      run: |
        if java Calculator 5 +; then
          echo "FAIL: expected non-zero exit code for missing argument"
          exit 1
        fi
        echo "PASS: missing argument correctly rejected"

  all-tests-passed:
    needs: [test, missing-args]
    if: always()
    runs-on: ubuntu-latest
    steps:
      - name: Check status of all test jobs
        run: |
          if [ "${{ needs.test.result }}" != "success" ] || [ "${{ needs.missing-args.result }}" != "success" ]; then
            echo "One or more test jobs failed."
            exit 1
          fi
          echo "All test jobs passed."
```

Output:


Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Calculator CI
Split CI workflow into build/test/summary jobs with matrix test cases #2
Re-run all jobs
Summary
Status
Success
Total duration
47s
Artifacts
1
ci.yml
Matrix: test
build
9 jobs completed
all-tests-passed
missing-args
Annotations
Usage
Workflow file


Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Calculator CI
Split CI workflow into build/test/summary jobs with matrix test cases #2
Re-run all jobs
Summary
Annotations
build
Search logs
Set up job
Checkout code
Set up JDK
Compile
Upload compiled class
Post Set up JDK
Post Checkout code
Complete job
Usage
Workflow file


Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Calculator CI
Split CI workflow into build/test/summary jobs with matrix test cases #2
Re-run all jobs
Summary
All jobs
build
test (addition, 3, +, 4, 7, false)
test (negative-operand, -5, +, 3, -2, false)
test (subtraction, 10, -, 4, 6, false)
test (multiplication, 6, *, 7, 42, false)
test (division, 10, /, 2, 5, false)
test (division-decimal, 7, /, 2, 3.5, false)
test (divide-by-zero, 5, /, 0, true)
test (invalid-operator, 5, %, 2, true)
test (non-numeric-operand, abc, +, 2, true)
missing-args
all-tests-passed
Run details
Usage
Workflow file
Run time
Learn about OS pricing on GitHub Actions
Job
Run time
build 10s
missing-args 8s
test (addition, 3, +, 4, 7, false) 7s
test (negative-operand, -5, +, 3, -2, false) 7s
test (subtraction, 10, -, 4, 6, false) 8s
test (multiplication, 6, *, 7, 42, false) 8s
test (divide-by-zero, 5, /, 0, true) 8s
test (division, 10, /, 2, 5, false) 9s
test (invalid-operator, 5, %, 2, true) 8s
test (division-decimal, 7, /, 2, 3.5, false) 4s
test (non-numeric-operand, abc, +, 2, true) 5s
all-tests-passed 3s
1m 25s

Lab 7 and 8

Aim:

Introduction to Docker & Docker Installation & Docker Commands & Dockerfile

Theory:

1. Introduction to Docker

Docker is a containerization platform used to package applications along with their dependencies into isolated environments called containers. It allows applications to run consistently across different computers, servers, and cloud environments.

2. Features of Docker

- **Portability:** Applications can run on different systems without major changes.
- **Isolation:** Containers run independently without interfering with each other.
- **Consistency:** Provides the same application environment across different machines.
- **Fast Deployment:** Containers can start quickly compared to virtual machines.
- **Easy Management:** Simplifies application deployment and maintenance.

3. Important Docker Concepts

- **Docker Image:** A packaged template containing an application and its required dependencies.
- **Docker Container:** An instance created from a Docker image that can execute an application.
- **Dockerfile:** A file containing instructions to build a Docker image.
- **Docker Hub:** An online platform used to store, share, and download Docker images.

- **Containerization:** The process of packaging applications and dependencies into isolated containers.

4. Difference Between Docker Image and Container

Docker Image:

1. Acts as a blueprint for creating containers.
2. Contains application code and dependencies.
3. Used to create multiple containers.

Docker Container:

1. Created from a Docker image.
2. Provides an isolated environment for applications.
3. Can be started, stopped, and removed.

5. Important Docker Commands

1. **docker pull mysql:** Downloads the MySQL image from Docker Hub.
2. **docker run mysql:** Creates and starts a container using the MySQL image.
3. **docker rm <container_id>:** Removes a stopped container without deleting its image.
4. **docker exec -it MYSQL /bin/bash:** Opens an interactive Bash terminal inside a running container named MYSQL.
5. **docker inspect <container>:** Displays detailed information about a container, including its configuration, network, IP address, and status.
6. **docker build -t image_name .:** Builds a Docker image using the Dockerfile in the current directory.
7. **docker push image_name:** Uploads a Docker image to Docker Hub.
8. **docker images:** Displays all locally available Docker images.
9. **docker ps:** Lists currently running containers.
10. **docker ps -a:** Lists all containers, including stopped containers.
11. **docker stop <container>:** Stops a running container.
12. **docker start <container>:** Starts an existing stopped container.
13. **docker rmi <image>:** Removes a Docker image.
14. **docker login:** Authenticates the user with Docker Hub.
15. **exit:** Exits the interactive terminal inside a container.

6. Important Dockerfile Instructions

1. FROM: Specifies the base image used to build the Docker image.

2. COPY: Copies application files into the Docker image.

3. CMD: Specifies the default command executed when the container starts.

7. Practical Tasks Performed

Task 1: Create a Project Folder

Created a folder named test to store the Python application and Dockerfile.

Task 2: Create a Python Program

Created a Python file named my_script.py containing a simple program to display "Hello from Docker!".

Task 3: Create a Dockerfile

Created a Dockerfile using Python 3.11 as the base image, copied the Python program, and specified the command to execute it.

Task 4: Build Docker Image

Built a Docker image using the Dockerfile and assigned it the name naitik002/python_file.

Task 5: Run Docker Container

Created and started a container using the built image.

The container executed the Python program and displayed "Hello from Docker!".

Task 6: Push Image to Docker Hub

Uploaded the created Docker image to Docker Hub for storage and sharing.

8. Docker Workflow

The general Docker workflow is:

1. Create an application.
2. Write a Dockerfile.
3. Build a Docker image.
4. Run a container using the image.
5. Verify the application output.
6. Push the image to Docker Hub.

Dockerfile → Build → Image → Run → Container

9. Advantages of Docker

1. Simplifies application deployment.
2. Reduces environment compatibility issues.
3. Provides lightweight application isolation.
4. Supports easy application sharing.
5. Improves development and testing efficiency.

Source Code:

Dockerfile

```
FROM python:3.11

COPY my_script.py .

CMD ["python", "my_script.py"]
```

Output:

Task-1

```
[naishal@Naishal ~ % cd ~/Desktop
[naishal@Naishal Desktop % mkdir test
[naishal@Naishal Desktop % cd test
[naishal@Naishal test % pwd
/Users/naishal/Desktop/test
```

Task-2

```
[naishal@Naishal test % nano my_script
[naishal@Naishal test % cat my_script
print("Hello from Docker!")
```

Task-3

```
[naishal@Naishal test % nano Dockerfile
[naishal@Naishal test % ls
Dockerfile      my_script.py
[naishal@Naishal test % cat Dockerfile
FROM python:3.11

COPY my_script.py .

CMD ["python", "my_script.py"]
naishal@Naishal test %
```

Task-4

```
hridaymulchandani@HridayM test % docker images
                                                                                i Info →   U In Use
IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
hriday002/python_file:latest         5199852d3dba       1.6GB         399MB
public.ecr.aws/lambda/python:3.12    6308f845cf3d        778MB         187MB
python:3.12-slim                     d764629ce0dd        179MB        45.4MB
hridaymulchandani@HridayM test %
```

```
[+] Building 372.9s (8/8) FINISHED                                docker:desktop-linux
 => [internal] load build definition from Dockerfile                               0.0s
 => => transferring dockerfile: 107B                                               0.0s
 => [internal] load metadata for docker.io/library/python:3.11                     7.3s
 => [auth] library/python:pull token for registry-1.docker.io                      0.0s
 => [internal] load .dockerignore                                                  0.0s
 => => transferring context: 2B                                                    0.0s
 => [internal] load build context                                                  0.0s
 => => transferring context: 68B                                                   0.0s
 => [1/2] FROM docker.io/library/python:3.11@sha256:701a80c1f60154d8e69          365.1s
 => => resolve docker.io/library/python:3.11@sha256:701a80c1f60154d8e6962          0.0s
 => => sha256:f66207ad7f472ea1fedd973717bae900544609ddca 6.19MB / 6.19MB          11.5s
 => => sha256:9f8f2b3eb1e3f4ccfcf8861267641428cb949a9f416f784 250B / 250B          0.7s
 => => sha256:c946ba9c36633c5ca599b3b506c9cb6abad07ef4 23.64MB / 23.64MB         80.3s
 => => sha256:0a48e1a423f7b0196a5eeb30fa9312d4ddac8 227.13MB / 227.13MB          362.6s
 => => sha256:fe8bb6fc1504f78dd21539a6470860dc388c329 67.63MB / 67.63MB          174.7s
 => => sha256:ea47142302fe74338cf6ed49f9110b011080a21c 25.04MB / 25.04MB         31.9s
 => => sha256:dfb68c024d5cdf4872bb7f0be8d63874aa6d0b7 49.78MB / 49.78MB          109.7s
 => => extracting sha256:dfb68c024d5cdf4872bb7f0be8d63874aa6d0b7a0050f1b0          0.5s
 => => extracting sha256:ea47142302fe74338cf6ed49f9110b011080a21c8ad97bad          0.2s
 => => extracting sha256:fe8bb6fc1504f78dd21539a6470860dc388c32916976117d          0.6s
 => => extracting sha256:0a48e1a423f7b0196a5eeb30fa9312d4ddac8315491bfd1d          1.9s
 => => extracting sha256:f66207ad7f472ea1fedd973717bae900544609ddcaf33075          0.1s
 => => extracting sha256:c946ba9c36633c5ca599b3b506c9cb6abad07ef454acf80c          0.5s
 => => extracting sha256:9f8f2b3eb1e3f4ccfcf8861267641428cb949a9f416f7845          0.0s
 => [2/2] COPY my_script.py .                                                      0.3s
 => exporting to image                                                             0.1s
 => => exporting layers                                                            0.0s
 => => exporting manifest sha256:4d1457b62e1b0594f859ebd12e25148981a4f50d          0.0s
 => => exporting config sha256:1310fa74a200c076afe710ff45819ba39f90299b32          0.0s
 => => exporting attestation manifest sha256:a1ee29232a9d8c51b2a856193b0d          0.0s
 => => exporting manifest list sha256:5199852d3dbabb65994791daba1eaec20d1          0.0s
 => => naming to docker.io/hriday002/python_file:latest                            0.0s
 => => unpacking to docker.io/hriday002/python_file:latest                         0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/dujd1auvtfq9wsm3vspb7o87c
```

Task-5

```
hridaymulchandani@HridayM test % docker run -it hriday002/python_file
Hello from Docker!
hridaymulchandani@HridayM test %
```

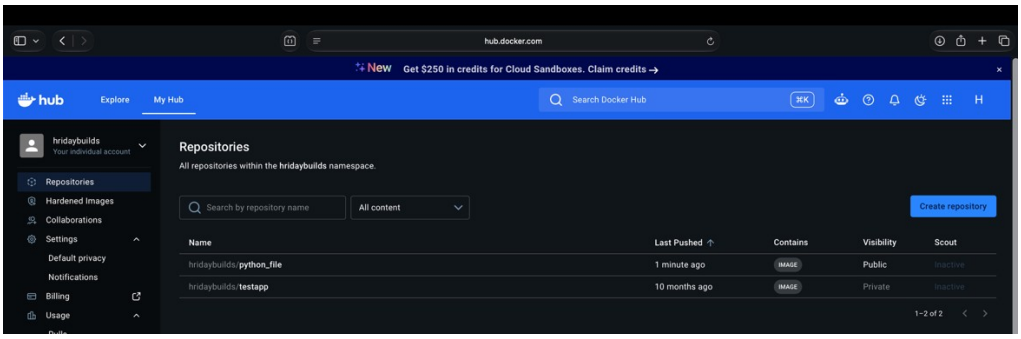
```
hridaymulchandani@HridayM test % docker ps -a
CONTAINER ID   IMAGE                  COMMAND                  CREATED          STATUS                      PORTS     NAMES
a98370e471bc   hriday002/python_file   "python my_script.py"   29 seconds ago   Exited (0) 29 seconds ago             frosty_greider
hridaymulchandani@HridayM test %
```

Task-6

```
hridaymulchandani@HridayM test % docker login
Authenticating with existing credentials... [Username: hridaybuilds]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
hridaymulchandani@HridayM test % docker tag hriday002/python_file:latest hridaybuilds/python_file:latest
hridaymulchandani@HridayM test % docker images
                                                                                                    i Info →  U  In Use
IMAGE                                      ID             DISK USAGE   CONTENT SIZE   EXTRA
hriday002/python_file:latest               5199852d3dba       1.6GB          399MB    U
hridaybuilds/python_file:latest            5199852d3dba       1.6GB          399MB    U
public.ecr.aws/lambda/python:3.12          6308f845cf3d        778MB          187MB
python:3.12-slim                           d764629ce0dd        179MB         45.4MB
hridaymulchandani@HridayM test %
```



Lab 9 and 10

Aim:

Testing using PyTest

Theory:

1. Introduction to PyTest

PyTest is a Python testing framework used to write and execute automated test cases. It helps developers verify code correctness, identify errors, and improve software reliability.

PyTest automatically detects test files and functions whose names begin with test_.

2. Features of PyTest

- **Automatic Test Discovery:** Automatically identifies and executes test functions.
- **Simple Testing:** Uses the assert statement to verify expected results.
- **Substring Matching:** Executes selected tests based on keywords in their names.
- **Markers:** Groups tests into categories for selective execution.
- **Fixtures:** Provides reusable setup data and resources for tests.
- **Test Reports:** Displays passed, failed, and skipped test cases.
- **CI Integration:** Supports automated testing in CI pipelines.

3. Important PyTest Concepts

- **Test Case:** A function that checks whether a particular part of the program works correctly.
- **Assertion:** A statement used to verify a condition. If the condition is false, the test fails.
- **Test Discovery:** PyTest automatically identifies files and functions following the test_ naming convention.

- **Marker:** A label assigned to test functions to organize and execute specific groups of tests.
- **Fixture:** A reusable function that provides required data or resources before test execution.
- **Parameterization:** Allows the same test to execute multiple times with different input values.

4. Practical Tasks Performed

Task 1: Basic Tests

Created a directory named pytest_demo and a test file named test_sample.py.

Three basic test functions were created:

1. test_greater() checks whether 100 is greater than 50.
2. test_greater_equal() checks whether 100 is greater than or equal to 100.
3. test_less() checks whether 100 is less than 200.

Command: pytest -v

This command executes all discovered tests and displays detailed results.

Task 2: Substring Matching

Substring matching allows developers to execute only those tests whose names contain a specific keyword.

Command: pytest -k great -v

This command executes the tests test_greater() and test_greater_equal() because their names contain the substring great.

It helps save time by running only relevant test cases.

Task 3: Grouping Tests Using Markers

Markers are used to categorize test functions into different groups.

Custom markers can be created using @pytest.mark.<markername>.

Command: pytest -m great -v

This command executes only the tests assigned the great marker.

Important Markers:

1. **skip:** Skips the execution of a test.
2. **xfail:** Marks a test as expected to fail.
3. **parametrize:** Runs the same test with multiple input values.

Task 4: Fixtures

Fixtures provide reusable data or setup operations required by test functions.

Created a file named test_fixture.py containing a fixture named input_value, which returns the value 10.

The test function test_addition() uses this value to verify that adding 5 produces 15.

Command: pytest test_fixture.py -v

This command executes the fixture based test and displays its result.

5. Important PyTest Commands

- **pytest:** Executes all discovered test cases.
- **pytest -v:** Executes tests with detailed output.
- **pytest -k great -v:** Runs tests containing the keyword great.
- **pytest -m great -v:** Runs tests assigned the great marker.
- **pytest test_fixture.py -v:** Executes tests from a specific file.

6. Advantages of PyTest

1. Simplifies automated testing in Python.
2. Reduces repetitive testing code.
3. Supports selective test execution.
4. Improves code reliability and debugging.
5. Provides clear and detailed test results.
6. Supports integration with CI pipelines.

Source Code:

test_task1_basic.py

```
def test_greater():
    num = 100
    assert num > 50


def test_greater_equal():
    num = 100
    assert num >= 100


def test_less():
    num = 100
    assert num < 200
```

test_task2_substring.py

```
def test_greater():
    num = 100
    assert num > 50
```

```
def test_greater_equal():
    num = 100
    assert num >= 100

def test_less():
    num = 100
    assert num < 200
```

test_task3_markers.py

```
import pytest

@pytest.mark.great
def test_greater():
    assert 100 > 50

@pytest.mark.great
def test_greater_equal():
    assert 100 >= 100

@pytest.mark.less
def test_less():
    assert 100 < 200
```

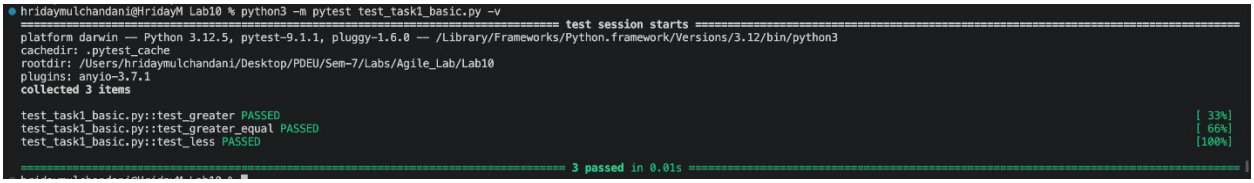
test_task4_fixture.py

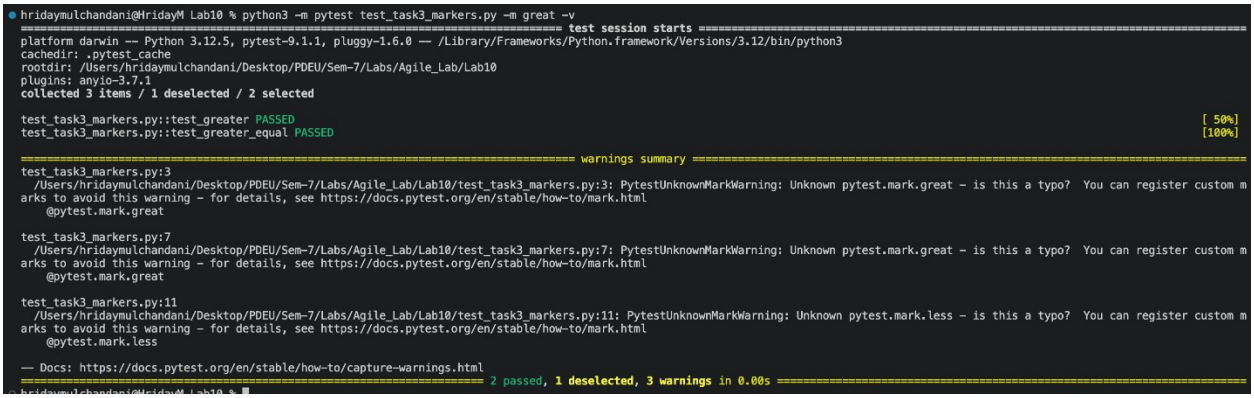
```
import pytest


@pytest.fixture
def input_value():
    return 10


def test_addition(input_value):
    assert input_value + 5 == 15
```

Output:





```
hridaymulchandani@HridayM Lab10 % python3 -m pytest test_task3_markers.py -m great -v
============================= test session starts ==============================
platform darwin -- Python 3.12.5, pytest-9.1.1, pluggy-1.6.0 -- /Library/Frameworks/Python.framework/Versions/3.12/bin/python3
cachedir: .pytest_cache
rootdir: /Users/hridaymulchandani/Desktop/PDEU/Sem-7/Labs/Agile_Lab/Lab10
plugins: anyio-3.7.1
collected 3 items / 1 deselected / 2 selected

test_task3_markers.py::test_greater PASSED                               [ 50%]
test_task3_markers.py::test_greater_equal PASSED                         [100%]

=============================== warnings summary ===============================
test_task3_markers.py:3
  /Users/hridaymulchandani/Desktop/PDEU/Sem-7/Labs/Agile_Lab/Lab10/test_task3_markers.py:3: PytestUnknownMarkWarning: Unknown pytest.mark.great - is this a typo?  You can register custom m
arks to avoid this warning - for details, see https://docs.pytest.org/en/stable/how-to/mark.html
    @pytest.mark.great

test_task3_markers.py:7
  /Users/hridaymulchandani/Desktop/PDEU/Sem-7/Labs/Agile_Lab/Lab10/test_task3_markers.py:7: PytestUnknownMarkWarning: Unknown pytest.mark.great - is this a typo?  You can register custom m
arks to avoid this warning - for details, see https://docs.pytest.org/en/stable/how-to/mark.html
    @pytest.mark.great

test_task3_markers.py:11
  /Users/hridaymulchandani/Desktop/PDEU/Sem-7/Labs/Agile_Lab/Lab10/test_task3_markers.py:11: PytestUnknownMarkWarning: Unknown pytest.mark.less - is this a typo?  You can register custom m
arks to avoid this warning - for details, see https://docs.pytest.org/en/stable/how-to/mark.html
    @pytest.mark.less

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
================== 2 passed, 1 deselected, 3 warnings in 0.00s ==================
```

```
hridaymulchandani@HridayM Lab10 % python3 -m pytest test_task4_fixture.py -v
============================= test session starts ==============================
platform darwin -- Python 3.12.5, pytest-9.1.1, pluggy-1.6.0 -- /Library/Frameworks/Python.framework/Versions/3.12/bin/python3
cachedir: .pytest_cache
rootdir: /Users/hridaymulchandani/Desktop/PDEU/Sem-7/Labs/Agile_Lab/Lab10
plugins: anyio-3.7.1
collected 1 item

test_task4_fixture.py::test_addition PASSED                              [100%]

============================== 1 passed in 0.01s ===============================
hridaymulchandani@HridayM Lab10 %
```

Lab 11

Aim:

Selenium tool

Theory:

1. Introduction to Selenium

Selenium is an open source web automation tool used to automate browser operations. It supports programming languages such as Python, Java, and C# and works with browsers like Chrome, Firefox, and Edge. Selenium is commonly used for automated web testing, browser interaction, and web scraping.

2. Features of Selenium

- **Browser Automation:** Automates actions such as opening websites, clicking buttons, and entering text.
- **Multiple Browsers:** Supports Chrome, Firefox, and Edge.
- **Multiple Languages:** Supports Python, Java, C#, and other languages.
- **Web Element Interaction:** Allows interaction with buttons, forms, and text fields.
- **Screenshot Capture:** Saves screenshots of webpages.
- **Automated Testing:** Reduces manual testing efforts.

3. Important Selenium Concepts

- **Selenium WebDriver:** A tool that controls web browsers through programming instructions.
- **Web Element:** An HTML component such as a button, textbox, or link present on a webpage.
- **Locator:** A method used to identify and access specific web elements.
- **Browser Automation:** Performing browser operations automatically using scripts.

- **Web Testing:** Checking whether website features work correctly through automated interactions.

4. Important Selenium Functions

- **driver.get(url):** Opens the specified webpage in the browser.
- **driver.find_element():** Locates a particular element on a webpage.
- **send_keys():** Enters text into input fields or sends keyboard inputs.
- **click():** Performs a click action on a web element.
- **driver.maximize_window():** Maximizes the browser window.
- **driver.save_screenshot():** Captures and saves a screenshot of the webpage.
- **driver.quit():** Closes the browser and ends the WebDriver session.

5. Methods to Locate Web Elements

- **ID:** Identifies an element using its unique ID.
- **Name:** Locates an element using its name attribute.
- **Class Name:** Identifies elements using their HTML class.
- **XPath:** Locates elements using their position or structure in the HTML document.
- **CSS Selector:** Identifies elements using CSS selection rules.

6. Practical Tasks Performed

Task 1: Create a Selenium Project

Created a project folder named selenium_demo and a Python file named test_selenium.py.

Task 2: Launch Web Browser

Imported Selenium WebDriver and launched the Google Chrome browser automatically.

Task 3: Open a Website

Used Selenium WebDriver to open the Google homepage and maximize the browser window.

Task 4: Locate Search Box

Located the Google search input field using the Name locator.

Task 5: Perform Search Operation

Entered the keyword "Selenium Python Tutorial" into the search box and pressed Enter to perform the search.

Task 6: Capture Screenshot

Captured and saved a screenshot of the search results page using the screenshot function.

Task 7: Close Browser

Closed the Chrome browser automatically after completing the operations.

7. Selenium Automation Workflow

1. Import Selenium WebDriver.
2. Launch the Chrome browser.
3. Open the required website.
4. Locate the desired web element.
5. Enter text or perform an action.
6. Capture the webpage screenshot.
7. Close the browser.

8. Advantages of Selenium

1. Reduces repetitive manual testing.
2. Supports multiple browsers and programming languages.
3. Automates website interactions.
4. Improves testing speed and efficiency.
5. Supports automated screenshot capture.
6. Helps identify errors in web applications.

Source Code:

test_selenium.py

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.common.keys import Keys
import time

driver = webdriver.Chrome()
driver.get("https://www.google.com")
driver.maximize_window()

search_box = driver.find_element(By.NAME, "q")
search_box.send_keys("Selenium Python Tutorial")
search_box.send_keys(Keys.RETURN)
```

```
time.sleep(3)

driver.save_screenshot("google_search_result.png")

driver.quit()
```

test_youtube.py

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.common.keys import Keys
from selenium.webdriver.chrome.options import Options
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
import time

options = Options()
options.add_experimental_option("detach", True)

driver = webdriver.Chrome(options=options)
driver.get("https://www.youtube.com")
driver.maximize_window()

wait = WebDriverWait(driver, 10)
search_box = wait.until(EC.element_to_be_clickable((By.NAME,
"search_query")))
```

```
search_box.send_keys("today's news")
search_box.send_keys(Keys.RETURN)

wait.until(EC.presence_of_element_located((By.ID, "video-title")))
time.sleep(2)

driver.save_screenshot("youtube_todays_news.png")

print("Page title:", driver.title)
print("Current URL:", driver.current_url)
```

test_youtube_video.py

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.common.keys import Keys
from selenium.webdriver.chrome.options import Options
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
import time

options = Options()
options.add_experimental_option("detach", True)
options.add_argument("--autoplay-policy=no-user-gesture-required")
```

```
driver = webdriver.Chrome(options=options)
driver.get("https://www.youtube.com")
driver.maximize_window()

wait = WebDriverWait(driver, 15)

search_box = wait.until(EC.element_to_be_clickable((By.NAME,
"search_query")))
search_box.send_keys("today's news")
search_box.send_keys(Keys.RETURN)

first_video = wait.until(EC.element_to_be_clickable(
   (By.CSS_SELECTOR, "ytd-video-renderer a#video-title")))
print("Opening video:", first_video.get_attribute("title"))
first_video.click()

wait.until(EC.presence_of_element_located((By.CSS_SELECTOR, "video")))
time.sleep(5)

try:
   skip_button = WebDriverWait(driver, 6).until(EC.element_to_be_clickable(
      (By.CSS_SELECTOR, ".ytp-skip-ad-button, .ytp-ad-skip-button")))
   skip_button.click()
   time.sleep(2)
   print("Ad skipped")
except Exception:
```

```
    print("No skippable ad found")

driver.execute_script("""
    var player = document.getElementById('movie_player');
    player.unMute();
    player.setVolume(100);
    player.setPlaybackRate(2);

    var video = document.querySelector('video');
    video.muted = false;
    video.volume = 1.0;
    video.playbackRate = 2;
""")

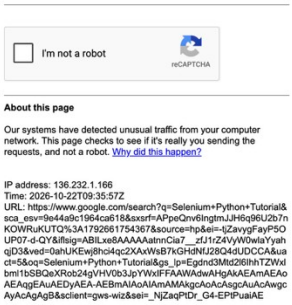
time.sleep(2)

speed = driver.execute_script("return
document.querySelector('video').playbackRate;")
volume = driver.execute_script("return document.querySelector('video').volume;")
print("Playback speed:", speed)
print("Volume:", volume)

driver.save_screenshot("youtube_video_2x.png")
print("Page title:", driver.title)
print("Current URL:", driver.current_url)
```

Output:


YouTube
today's news
Sign in
Top chat replay
Top fans
जनहित
Live chat replay is on. Messages that appeared when the stream was live will show up here.
CJP Protest News LIVE: सुबह-सुबह आई CJP पर बड़ी खबर सामने | Saurav Das | Ashutosh Ranka | HC Hearing
ABP NEWS
51.4M subscribers
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68
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New
LIVE Do Took: कोर्ट की फटकार, रीजीम चेंज गैंग में हाहाकार! | Sanjay Ra...


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Our systems have detected unusual traffic from your computer network. This page checks to see if it's really you sending the requests, and not a robot. Why did this happen?
IP address: 136.232.1.166
Time: 2026-10-22T09:35:57Z
URL: https://www.google.com/search?q=Selenium+Python+Tutorial&sca_esv=9e44a9c1964ca618&sxsrf=APpeQnv6IngtmJJH6q96U2b7nKOWRuKUTQ%3A1792661754367&source=hp&ei=-tjZavygFayP5OUP07-d-QY&iflsig=ABILxe8AAAAAatnnCla7__zfJ1rZ4VyW0wlaYyahqjD3&ved=0ahUKEwj8hci4qc2XAxWsB7kGHdNfJ28Q4dUDCCA&uact=5&oq=Selenium+Python+Tutorial&gs_lp=Egdnd3Mtd2l6IhhTZWxlbml1bSBQeXRob24gVHV0b3JpYWxlIFAAWAdwAHgAkAEAmAEAoAEAqgEAuAEDyAEA-AEBmAIAoAIAmAMAkgcAoAcAsgcAuAcAwgcAyAcAgAgB&sclient=gws-wiz&ei=_NjZaqPtDr_G4-EPlPuaiAE